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To Preserve, or Not to Preserve Volume, that is the Question - for Normalizing Flows

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Abstract

English Original / Englisches Original

Normalizing flows are a promising candidate to model complex high-dimensional densities. They can be categorized into *volume-preserving* (VP) and *non-volume preserving* (NVP). We investigate in which scenarios one or the other should be preferred.

We examine the difference between VP and NVP normalizing flows by introducing the *Density of States* (DoS) to the field of deep generative models, which describes the distribution of log density. The ability to change the volume leads to the ability to change the DoS, thus NVP normalizing flows can adjust the DoS, while VP normalizing flows cannot. Matching the DoS is a necessary condition to approximate a distribution well, and so the DoS of the normalizing flow and of the distribution must match. We derive that when a manifold is *inflated* with noise that matches the prior of the normalizing flow, and the ratio between the manifold dimensionality d and the data dimensionality D is small, the fixed DoS of VP normalizing flows and the DoS of the the inflated manifold can be matched. Thus, for small $\frac{d}{D}$ and prior matching noise, VP and NVP normalizing flows can perform comparable for the approximation of the distribution, and vice versa: when $\frac{d}{D}$ is large or the inflation noise does not match the distribution, VP normalizing flows should not be used.

We experimentally validate our theoretical findings, and find that NVP normalizing flows often perform worse than VPs on the approximation of the distribution of the manifold for small $\frac{d}{D}$. We investigate the reason for this and find that the term which allows NVP normalizing flow to change the volume is culprit, because it causes a steep loss space that leads to an unstable training, which leads to convergence in bad local minima.

Abstrakt

German Translation / Deutsche Übersetzung

Normalizing Flows sind ein vielversprechender Kandidat für die Modellierung komplexer hochdimensionaler Verteilungen. Sie können in *volumen-erhaltende* (VP) und *nicht-volumen-erhaltende* (NVP) Normalizing Flows unterteilt werden. Wir untersuchen in welchen Szenarien das eine oder das andere bevorzugt werden sollte.

Wir untersuchen den Unterschied zwischen VP und NVP Normalizing Flows, indem wir die *Zustandsdichte* (DoS) (zu engl. *Density of States*) in den Bereich der deep generative Models ein. Die DoS beschreibt die Verteilung der Log-Dichte. Die Möglichkeit, das Volumen zu ändern, führt zu der Möglichkeit die DoS zu ändern, somit können NVP Normalizing Flows die DoS anpassen, während VP Normalizing Flows dies nicht können. Die Anpassung der DoS ist eine notwendige Bedingung für eine gute Annäherung an eine Verteilung. Deshalb muss die DoS des Normalizing Flows und die der Verteilung übereinstimmen. Wir leiten her, dass, wenn eine Mannigfaltikeit mit Rauschen *aufgeblähten* (zu engl. *inflated*) wird, so dass die Verteilung des Rauschens mit dem der Verteilung des Priors des Normalizing Flows übereinstimmt, und das Verhältnis zwischen der Dimensionalität der Mannigfaltikeit d und der Dimensionalität der Daten D klein ist, die fixierte DoS der VP Normalizing Flows und die DoS der aufgeblähten Mannigfaltikeit übereinstimmen. Daher gilt für kleine $\frac{d}{D}$, dass VP und NVP Normalizing Flows bei der Annäherung an die Verteilung vergleichbare Leistungen erbringen können, und dass NVP Normalizing Flows bevorzugt werden sollten, wenn $\frac{d}{D}$ groß ist.

Wir validieren unsere theoretischen Erkenntnisse experimentell und stellen fest, dass VP Normalizing Flows die Verteilung der Mannigfaltigkeit oft besser annähern als NVP Normalizing Flows für kleine $\frac{d}{D}$. Wir untersuchen den Grund dafür und finden heraus, dass der Term, der es dem NVP Normalizing Flows erlaubt, das Volumen zu verändern, verantwortlich für die schlechte Leistung von NVP Normalizing Flows ist, da dieser Term einen steilen Loss Space verursacht, der zu einem instabilen Training führt, welches wiederum zu einer Konvergenz in schlechte lokale Minima führt.

Statement of Authorship

I hereby declare that the thesis submitted is my own, unaided work, completed without any unpermitted external help. Only the sources and resources listed were used.

The independent and unaided completion of the thesis is affirmed by affidavit:

Ihringen,	December 31, 2022	D. Luibrand
Location,	Date	Signature

Contents

Nomenclature

In this section we provide a table with frequently used notation and symbols.

A	Capital letters denote random variables.
$\mathcal{A}$	Curved capital letters denote the space of A.
a	Small letters denote a sample of their respective random variable.
p_A	Distribution of random variable A.
$p_A(\cdot)$	Probability density function of random variable A.
$p(a)$	Probability density function evaluated at position a,
	where the distribution of a is given by the context (usually p_A).
D	Dimensionality of the inflated distribution and of the normalizing flow.
d	Dimensionality of the manifold.
p_Z	Base distribution of a normalizing flow, D dimensional.
p_X	Target distribution of inflated data, D dimensional.
q_X	Learned distribution, D dimensional.
p^*	True distribution, context-dependent dimension.
p_M	True distribution of the manifold, d dimensional.
q_M	Learned distribution for the manifold, d dimensional.
$\mathcal{N}(\mu, \sigma^2)$	Gaussian distribution with mean μ and variance σ^2.
$\mathcal{U}[a, b]$	Uniform distribution between a and b, for $a, b \in \mathbb{R}$.
$f : \mathcal{X} \to \mathcal{Z}$	Transformation from the data to the latent space.
$v_f(x) = \left\lvert \det \frac{\partial f}{\partial x} \right\rvert$	Volume element, respectively determinant of the Jacobian of f
	evaluated at position x.

Glossary

pdf	Probability density function
DoS	Density of states
Degenerate-DoS distribution	Distribution with a DoS that has its mass at a single point
Non-degenerate-DoS distribution	Distribution that is not a degenerate-DoS distribution
Inflated manifold	Manifold with noise added in the normal space of the manifold
NVP	Non-volume preserving (normalizing flow)
VP	Volume preserving (normalizing flow)
NICE	A VP normalizing flow [1]
RealNVP	An NVP normalizing flow [2]
RealVP	RealNVP but with volume preserving transformations
RealNC	RealNVP but the volume element is replaced by a normalizing constant
VP models	The VP normalizing flows we use for our experiments, i.e. NICE and RealVP
NVP models	The NVP normalizing flow we use for our experiments, i.e. RealNVP
Large $\frac{d}{D}$	When $d = 1$ and $D = 2$
Small $\frac{d}{D}$	When $d = 1$ and $D = 36, 100, 196$
Regime	A specific $\frac{d}{D}$ (i.e. large or small)

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Chapter 1

Introduction

1.1 Motivation

The ability to model the generative process of observed samples is an important task. Applications for such models include density estimation, inference, outlier detection and data generation. Since data can become very complex, such models should be highly expressive, and therefore deep neural networks are promising building blocks for (*deep*) generative models. Several families of deep generative models have been developed to accomplish these tasks, such as diffusion models [3], GANs [4], VAEs [5], autoregressive models [6] and normalizing flows [7, 8]. We focus on normalizing flows which are popular because, unlike GANs, VAEs and diffusion models they can provide exact likelihoods for given samples, and unlike autoregressive models, they can perform sampling and likelihood calculation in a computationally efficient manner. This is achieved by a bijective and differentiable function $f : \mathcal{X} \to \mathcal{Z}$ that transforms the data space $\mathcal{X}$ with a complex unknown distribution p_X into a latent space $\mathcal{Z}$ with a simple base distribution p_Z. With the bijective map, the density evaluated at position $x \in \mathcal{X}$ can be computed using the change of variables formula:

$$p_X(x) = p_Z(f(x)) \left|\det \frac{\partial f(x)}{\partial x}\right| = p_Z(f(x))\, v_f(x), \tag{1.1}$$

where $\left|\det \frac{\partial f(x)}{\partial x}\right| = v_f(x)$ is the determinant of the Jacobian of f at position x, which is referred to as *volume element* [9].

Normalizing flows have shown impressive performance in image generation [10, 11], audio generation [12] and audio manipulation [13], physics [14, 15], and reinforcement learning [16, 17]. Most state-of-the-art normalizing flows are *non-volume preserving* (NVP), which means that the transformation f can change the volume $v_f(x)$ depending on x, for instance by using affine transformations [2, 10, 11]. The predecessors of NVP normalizing flows are *volume-preserving* (VP) normalizing flows, e.g. NICE [1], which keep the volume constant, i.e. $v_f(x) = c,\ \forall x \in \mathcal{X}$. VP and NVP transformations are illustrated in Figure 1.2. NVP transformations are clearly more expressive than VP transformations.

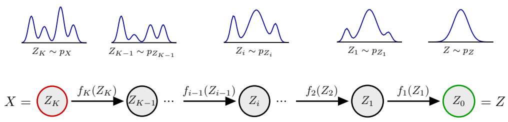

Figure 1.1: Series of invertible transformations from the latent space with a known simple distribution p_Z into the data space of a complex distribution p_X.[1]

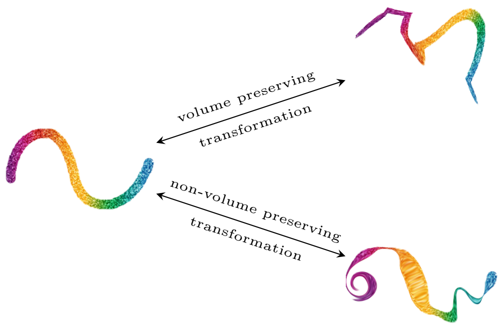

Figure 1.2: A VP and an NVP transformation. The color indicates where a respective point was mapped to. NVP transformations can stretch and compress areas, while VP transformations cannot.

However, recent studies [9, 18] revealed an error mode of NVP normalizing flows: under some situations they assign higher likelihoods to out-of-distribution data than to in-distribution data. Nalisnick et al. [9] focus on NVP normalizing flows and showed that this effect can be attributed to the inaccurately learned volume element. However, they also reported that VP normalizing flows show similar behavior. Although such interesting findings have been reported, there is not much investigation on the importance and difficulties of using the volume change, and therefore whether we should use VP or NVP models under what situations. Considering that NVP normalizing flows have a larger function space and that Occam's razor has been shown to produce enhanced Bayesian models [19], VP normalizing flows may perform better when a volume change is not needed.

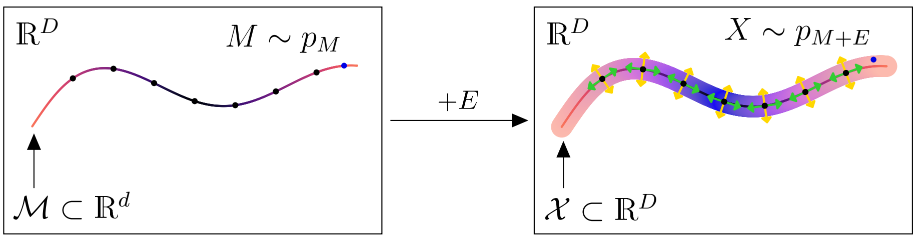

Figure 1.3: The color gradients along the lines illustrate the pdf of the respective distribution. The black dots depict the given samples. They are distributed according to p_M. Adding noise E to the samples dequantizes them (yellow and green arrows) and inflates the dimensionality of the manifold $\mathcal{M}$ (yellow arrows). The resulting distribution is p_{M+E}.

Another issue of normalizing flows is that they learn a distribution in the input space with dimensionality D, but natural data tends to reside on a smaller, d ($< D$), dimensional manifold [20] (see left side of Figure 1.3.). Even when $d = D$, we typically learn from a finite set of samples which could be learned by placing high density spikes on each sample [21]. A widely used method to circumvent this is to add noise to the samples [1, 2, 10, 21, 22, 23, 24, 25, 26, 27]. This *dequantizes* the samples [2, 21] and *inflates* the dimensionality [23] (see Figure 1.3). We refer to this process as

[1] Figure inspired by `https://github.com/janosh/awesome-normalizing-flows`.

inflation. The density estimator is then trained and evaluated on the inflated distribution. Since the true non-inflated distribution of the manifold is usually unknown, only a few studies [23, 27, 28] check whether the manifold distribution has been properly approximated. None of these studies examined whether VP or NVP normalizing flows better approximates the manifold, nor did they examine how adding noise affects the need to be able to change the volume.

In this thesis we will theoretically analyze under which conditions a volume change is necessary and when it can be preserved to approximate a distribution well. In particular, we will focus on dimensionality and the inflation noise type. To validate our theoretical results empirically, we will train VP and NVP normalizing flows on manifolds with different data space dimensionality and different types of inflation noise and measure how well they approximate those distributions. We will use manifolds with known ground truth topology and distribution, which allows us to measure if VP or NVP normalizing flows can approximate the manifold distribution better. Furthermore, we will conduct exploratory experiments to investigate other differences that arise from the choice of VP or NVP transformations.

1.2 Approach

We introduce the *Density of States* (DoS) to the field of deep generative models, which describes the distribution of log density. This concept originates from physics, where it describes the amount of states for a given energy in a system [29]. We use the DoS to examin the difference between VP and NVP normalizing flows. We find that VP normalizing flows preserve the DoS up to a fixed translation, which corresponds to the constant volume change, while NVP normalizing flows can adjust it. Note that while VP normalizing flows cannot change the DoS of the whole sample space, they can change the DoS of a subset of it, e.g. the manifold.

Using this notion, we show that volume change becomes less necessary when $\frac{d}{D}$ is small, and vice versa. We validate this theoretical finding empirically by evaluating how well VP and NVP normalizing flows approximate distributions of different inflated manifolds and different embedding spaces dimensionalities. To show that this is not just an effect of the different types of transformations, i.e. additive for VP and affine for NVP normalizing flows, we introduce VP affine transformations and observe consistent results.

We use manifolds with known ground truth topology and distribution and can thus measure how well the models approximate the density along the manifold. Consistent with the approximation of the inflated manifold NVP normalizing flows perform well for large $\frac{d}{D}$, while VP normalizing flows do not. On the other hand, for small $\frac{d}{D}$, NVP normalizing flows perform poorly in approximating the density of the manifold, while VP normalizing flows perform acceptably.

In the general problem setting, we can only measure how well a model approximates an inflated manifold using the likelihood of a given test set, but we cannot measure how well the (deflated) manifold is approximated[1]. However, if we are interested in good likelihoods for unseen in-distribution samples of the manifold, which meaning that the (deflated) manifold should be well approximated. Since we must rely on the likelihood for the inflated distribution to predict a good

[1]We could measure good sampling quality, but it has been shown that good sample quality is not equivalent to good density estimation [25].

approximation of the manifold, and since we know the true manifold topology and its distribution, we evaluate whether a good approximation of the inflated manifold predicts a good approximation of the manifold density. We find that for large $\frac{d}{D}$, the likelihood for the inflated manifold of VP normalizing flows poorly predicts a good approximation of the manifold density, while it predicts well the approximation of the manifold density for NVP normalizing flow, and vice versa for small $\frac{d}{D}$.

Since NVP normalizing flows are more expressive than VP normalizing flows, one might expect that they should perform at least comparable. However, they show poor performances in approximating the manifold for small $\frac{d}{D}$. We investigate this empirically and show that the volume element is responsible. This is consistent with the findings of Nalisnick et al. [9], who find the same responsibility for the poor performance on out-of-distribution data. We find that the loss space of NVP normalizing flows is more peaky than that of VP normalizing flows, which manifests itself in strong weight fluctuations during training. We conclude that the peakiness of the loss space could be the reason for the poor performance we see in NVP normalizing flows.

From the notion we introduce, we also derive that when using VP normalizing flows, uniform noise should not be used, but noise that matches the distribution of the prior of the normalizing flow. We validate this theoretical finding empirically. We note that existing comparisons between VP and NVP normalizing flows use uniform noise inflation, e.g. Kingma & Dhariwal [11, Figure 3] find that the NVP version of their model approximates a distribution better than the VP version, and Nalisnick et al. [9, Figure 3 and 4] find that both VP and NVP normalizing flows assign higher likelihood to out-of-distribution data[1]. Thus, their results might have been different if prior matching inflation noise would have been used.

Finally, we visually examine the learned transformation of VP and NVP normalizing flows in two-dimensional space and find that both models transform the manifold from the data space to the latent space by applying a zigzag pattern to the manifold to fit it in the high probability region, i.e. the center, of the latent distribution. This explains the ups and downs we observe in the learned pdfs along the manifold.

1.3 Thesis Outline

Chapter 1 motivates our research and gives an overview over the achieved results. Chapter 2 explains normalizing flows, other relevant notions and introduces volume preserving affine transformations. Chapter 3 theoretically analyzes the density of states of arbitrary distributions and normalizing flows. In Chapter 4 we experimentally confirm our theoretical findings and explore the poor performance of NVP normalizing flows. The thesis ends with a conclusion and an outlook in Chapter 5.

[1] Both papers use the Glow model, which is trained on uniform noise as can be derived from the code: `https://github.com/openai/glow/blob/91b2c577a5c110b2b38761fc56d81f7d87f077c1/model.py#L171`.

Chapter 2

Background and Related Work

This chapter starts with the general learning problem, then introduces normalizing flows along with our proposed normalizing flow. We then provide insights into current research questions for normalizing flows, and finally state some of the methods we use in our experiments.

2.1 Problem Setting

Let $M \sim p_M$ be a continuous random variable from a d-dimensional manifold $\mathcal{M}$ embedded in a D-dimensional space $\mathcal{X}$. Given a set of training samples $\{m_i\} = \mathcal{D}$, where m_i are samples drawn from p_M, we are interested in approximating the probability density function (pdf) $p_M(\cdot)$ and a way to generate samples from p_M. We use a model class $\{q_\theta \,|\, \theta \in \Theta\}$ and minimize the Kullback–Leibler divergence from q_θ to p_M to approximate p_M.

Definition 2.1. (Kullback–Leibler divergence [30, Equation 2.4])
Given two continuous distributions p and q, the Kullback–Leibler (KL) divergence from q to p is

$$\mathrm{D}_{\mathrm{KL}}(p(x) \;\|\; q(x)) = \int p(x) \log \frac{p(x)}{q(x)}\, dx \approx -\frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \log q(x) - \mathbb{H}(p), \tag{2.1}$$

where $\mathbb{H}(p)$ is the entropy of p.

Since $\mathbb{H}(p_M)$ is a constant quantity, minimizing the KL divergence results in maximizing $\frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \log q_\theta(x)$, which is known as maximum log-likelihood [31]. In this work, our model class is normalizing flows.

2.2 Normalizing Flows

Definition 2.2. (Diffeomorphism [[32, Definition 1.1.9]]) A function f is called a diffeomorphism if its inverse f^{-1} exists and f and f^{-1} are differentiable.

In this section we introduce normalizing flows along with architectural decisions for them. Normalizing flows are a family of deep generative models that use diffeomorphisms to transform the data space $\mathcal{X} = \mathbb{R}^D$ with a complex unknown probability distribution p_X into a space $\mathcal{Z}$ with a simple latent base distribution p_Z. They were proposed by Tabak and Vanden-Eijnden [8] in 2010 and popularized in the machine learning community by Rezende and Mohamed [7] in 2015. Normalizing flow achieve exact inference by using the change of variables formula 2.2 and a function composition $f = f_1 \circ ... \circ f_K : \mathcal{X} \to \mathcal{Z}$ of invertible transformations $f_1, ..., f_K : \mathbb{R}^D \to \mathbb{R}^D$,

$$p(x) = p(z_K) = p_z(z_0) \left| \det \frac{\partial f(x)}{\partial x} \right| = p_z(z_0) \prod_{i=1}^{K} \left| \det \frac{\partial f_i(z_i)}{\partial z_i} \right| = p_z(z_0)\, v_f(x), \qquad \text{with } z_{i-1} = f_i(z_i), \tag{2.2}$$

where the base distribution p_Z is usually an isotropic Gaussian and $v_f(x)$ is the volume change (also known as Jacobian of f) at position x. To efficiently compute the density at a position x, the Jacobian of f_i needs to be easily computable, which is ensured by the construction of f_i explained in the upcoming subsection. To generate samples from a normalizing flow, one takes samples from the base distribution and applies the inverse transformation $f^{-1} = f_K^{-1} \circ ... \circ f_1^{-1}$. Invertibility of f_i can be provided by *coupling layers* [1], autoregressive layers [6], invertible linear layers [11] or residual layers [33]. Since coupling layers are fast, widely used in research [9, 18, 34, 35, 36, 37], and have been proven to be universal diffeomorphism approximators [38], we restrict our research to them. For a full review on normalizing flows, see [24, 39]. Note that in this work we consider the direction of flow from the data to the latent space.

2.2.1 Coupling Layer

In this section we explain coupling layer, which are the main building blocks or normalizing flows. Let f_i be a coupling layer with input $x = (x_1, ..., x_D)$. The input is partitioned into $x_{\text{id}} = x_{I_1}$ and $x_{\text{change}} = x_{I_2}$, where I_1, I_2 is a partitioning of $\{1, ..., D\}$. Partitioning x_{id} will not be changed by the coupling layer, it is processed by a function g of arbitrary output and arbitrary complexity. The main transformation within the coupling layer is made by a diffeomorphic function $h : \mathbb{R}^{|I_2|} \times g(\mathbb{R}^{|I_1|}) \to \mathbb{R}^{|I_2|}$. The transformation of f_i is

$$f_i(x) = \begin{pmatrix} x_{\text{id}} \\ h(x_{\text{change}};\, g(x_{\text{id}})) \end{pmatrix} = \begin{pmatrix} y_{I_1} \\ y_{I_2} \end{pmatrix} = y. \tag{2.3}$$

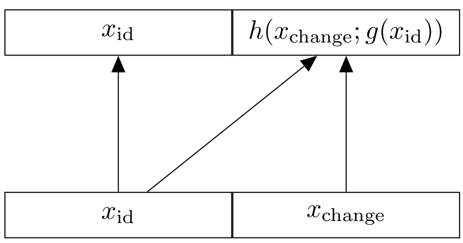

Figure 2.1: Computational graph of a coupling layer.

The inverse direction is

$$f_i^{-1}(y) = \begin{pmatrix} y_{I_1} \\ y_{I_2} \end{pmatrix} = \begin{pmatrix} x_{\text{id}} \\ h^{-1}(y_{I_2};\, g(y_{I_1})) \end{pmatrix}, \tag{2.4}$$

and the Jacobian of f_i is of form

$$\frac{\partial f_i}{\partial x} = \begin{pmatrix} \mathbb{1}_{|I_1|} & 0 \\ \frac{\partial y_{I_2}}{\partial x_{\text{id}}} & \frac{\partial y_{I_2}}{\partial x_{\text{change}}} \end{pmatrix}, \tag{2.5}$$

which implies that $\det \frac{\partial f_i}{\partial x} = \det \frac{\partial y_{I_2}}{\partial x_{\text{change}}}$.

Several methods have been proposed for partitioning, such as odd and even indices [1] (also known as checkerboard pattern), horizontal pattern [34], or a channel-wise pattern [2]. g is implemented as a neural network, whose architecture is usually chosen depending on the data type, such as convolutional neural networks for images [1, 2, 11] or fully connected neural networks for tabular data [34]. Multiple transformations for h have been proposed, such as *additive coupling layers* [1], *affine coupling layers* [2], and different types of *spline layers* [40, 41, 42]. For NVP transformations, we use affine coupling layers since they are widely used in the literature, e.g. [9, 34, 35, 43]. For VP transformations, we use additive coupling layers [1], which have been used as counterparts to affine coupling layer [11]. To eliminate the possibility that our results are simply due to the difference in expressiveness of the transformations (i.e. NVP transformations are more expressive than VP transformations), we introduce VP affine coupling layers in Section 2.2.3.

Affine coupling layers [2, Equation 5] choose

$$g_{st} : \mathbb{R}^{|I_1|} \to \mathbb{R}^{|I_2|} \times \mathbb{R}^{|I_2|}, \quad g_{st}(x_{\text{id}}) = (\, s(x_{\text{id}}),\ t(x_{\text{id}})\,) \tag{2.6}$$

$$h_{st}(x_{\text{change}};\ g_{st}(x_{\text{id}})) = (x_{\text{change}} + t(x_{\text{id}})) \odot \exp(\, s(x_{\text{id}})\,), \tag{2.7}$$

where $\odot$ is the Hadamard product [44, Definition 1.1], s is the scaling net, and t the translation net. The log volume element for Equation 2.2 and 2.5 can be efficiently computed as

$$\log\left|\det \frac{\partial f_i}{\partial x}\right| = \log\left|\prod_{i=0}^{|I_1|} 1 \cdot \prod_{i=0}^{|I_2|} \exp(s_i(x_{\text{id}}))\right| = \sum_{i=1}^{|I_2|} s_i(x_{\text{id}})\,. \tag{2.8}$$

Additive coupling layers [1, Section 3.2] choose

$$g_t : \mathbb{R}^{|I_1|} \to \mathbb{R}^{|I_2|}, \quad g_t(x_{\text{id}}) = t(x_{\text{id}}) \tag{2.9}$$

$$h_t(x_{\text{change}};\ g_t(x_{\text{id}})) = x_{\text{change}} + t(x_{\text{id}}). \tag{2.10}$$

This has a determinant of 1.

2.2.2 Scale Layer

To account for some volume change, the last layer of VP normalizing flows is a scale layer s_{const} with learnable scalings θ_s [1].

Scale layer [1, Section 3.3] transform an input x as

$$s_{\text{const}}(x) = (x \odot \theta_s). \tag{2.11}$$

Which has a volume element of $\left|\prod_{s\in\theta_s} s\right|$.

2.2.3 Volume Preserving Affine Coupling Layers - RealVP

In this section, we introduce a VP version of affine coupling layers. The function g has the same form as standard affine coupling layers, but the diffeomorphism h becomes volume preserving.

Volume preserving affine coupling layers choose

$$g_{vp\text{-}st} : \mathbb{R}^{|I_1|} \to \mathbb{R}^{|I_2|} \times \mathbb{R}^{|I_2|}, \quad g_{vp\text{-}st}(x_{\text{id}}) = (\, s(x_{\text{id}}),\ t(x_{\text{id}})\,) \tag{2.12}$$

$$h_{vp\text{-}st}(x_{\text{change}};\ g_{vp\text{-}st}(x_{\text{id}})) = (x_{\text{change}} + t(x_{\text{id}})) \odot \frac{\exp(s(x_{\text{id}}))}{\nu(x_{\text{id}})} \tag{2.13}$$

$$\nu : \mathbb{R}^{|I_2|} \to \mathbb{R}, \quad \nu(x_{\text{id}}) = \left(\prod_{i=1}^{|I_2|} \exp(s_i(x_{\text{id}})) \right)^{\frac{1}{|I_2|}} . \tag{2.14}$$

The function ν normalizes the volume change such that the volume element becomes

$$\left| \prod_{i=0}^{|I_2|} \frac{\exp(s_i(x_{\text{id}}))}{\nu(x_{\text{id}})} \right| = \left| \frac{\prod_{i=0}^{|I_2|} \exp(s_i(x_{\text{id}}))}{\nu(x_{\text{id}})^{|I_2|}} \right| = \left| \frac{\prod_{i=0}^{|I_2|} \exp(s_i(x_{\text{id}}))}{\left(\prod_{i=0}^{|I_2|} \exp(s_i(x_{\text{id}})) \right)^{\frac{|I_2|}{|I_2|}}} \right| = 1\,, \tag{2.15}$$

which makes it volume preserving. In practice, $\frac{\exp(s(x_{\text{id}}))}{\nu(x_{\text{id}})}$ is computed in log space to achieve numerical precision.

We refer to a model with VP affine coupling layers as *RealVP* in conjunction with its NVP counterpart RealNVP [2]. We equip this model with a final scale layer (see Section 2.2.2) to account for some volume change. A proof that RealVP is more expressive than NICE is given in Appendix A.2. Note that when the data dimensionality D is 2, VP affine coupling layers coincide with additive coupling layers[1].

2.3 Inflation, Dimensionality and the Manifold

In this section we introduce open research questions of normalizing flows Standard normalizing flows learn a distribution of dimensionality D which is determined by the input space. However, according to the manifold hypothesis [20], natural data tend to lie on a manifold with dimensionality d ($< D$). This implies that the distribution is expressed by a Dirac delta function, which cannot be learned. Even when $d = D$, we typically learn from a finite set of samples that could be learned by placing high density spikes on each sample [21].

The most widely used method to circumvent this problem is to add noise E to the samples [1, 2, 10, 21, 22, 23, 24, 25, 26, 27], thereby dequantizing the samples [2, 21] and inflating the dimensionality [23]. The distribution to be learned becomes p_{M+E}, which we will refer to as *inflated distribution*. It is an open research question how E should be chosen. Dinh et al. [1, 2] and Uria et al [21] choose uniform noise $\mathcal{U}[-a, a]^D$, $a \in \mathbb{R}^+$. Since these partial hypercubes[2] around each sample are difficult to learn for smooth function approximators, Ho et al. [10] propose to simultaneously learn a conditional flow based generative model $q_{E|M}$ to generate more flexible noise.

[1] $|I_1| = |I_2| = 1$, thus the scaling $\frac{\exp(s(x_{\text{id}}))}{\nu(x_{\text{id}})}$ will resolve to 1.

[2] Two samples that reside close to each other on the manifold have overlapping hypercubes.

Together with other architectural changes, they achieve better results on the inflated distribution as their predecessors.

Given that the manifold is not too entangled and not too curved, and that its dimensionality is known, the *inflation-deflation* approach of Horvat and Pfister [23] adds Gaussian noise in the normal space of the manifold. Since this makes the noise independent of the manifold, the pdf is of the form $p(m + e) = p_M(m)\, p(e)$, which leads to a deflation term for the learned density q_{M+E} that yields the pdf for the learned manifold, i.e. $q_M(m) = \frac{q_{M+E}(m)}{p_E(0)}$, assuming the noise is centered at 0. They evaluate their approach on a fixed manifold in different dimensions inflated with different noise strength, and find that with optimal noise, a model can perform equal on 10 and 20 dimensions, and that too small ($\sigma^2 \lessapprox 10^{-4}$) and too high ($\sigma^2 \gtrapprox 10^{-1}$) noise variances lead to poor performance along the manifold [23, Figure 4].

Although the ultimate goal is to learn the distribution of the manifold, the aforementioned methods cannot separate the manifold from the inflated manifold (i.e. dimensionality reduction), nor can they directly measure it directly, nor can they generate samples from it. To overcome this Horvat and Pfister [27] propose an architecture consisting of a denoising normalizing flow, which has the objective of a denoising autoencoder [45], followed by a projection of the noise-insensitive part (i.e. the manifold) and a projection of the noise-sensitive part (i.e. the inflation noise). The noise-insensitive part is processed by a consecutive normalizing flow, and the noise-sensitive is measured using the pdf of the added noise. Similar methods have been proposed [28, 46, 47].

Since diffusion probabilistic models (DPM) [3] have gained much attention due to impressive results in image generation [48], their relation to this topic is briefly presented in simple terms. To train a DPM, one successively adds isotropic normal noise to the manifold in the full embedding space until a standard normal distribution in the dimensionality of the manifold (D) remains. Then a model is trained to reverse this process. In other words the generative process of DPMs is trained to undo the noise which is added. Since we consider the difference in volume and non-volume-preserving, we mention that DPMs are NVP. This can be seen since their pushforward distribution is $\mathcal{N}(x|f(z))$, where f is a non-linear transformation of the latent variable $z \sim \mathcal{N}^D(0, 1)$. Due to the complexity of f, $\mathcal{N}(x|f(z))$ is not normally distributed and thus DPMs are not volume preserving.

2.4 Affine vs. Additive Transformations

Additive coupling layers are a proper subset of affine coupling layers (for $s(\cdot) = 0$, see Equation 2.7 and 2.10), thus affine transformations are more expressive and should outperform additive transformations. However, little research has been done to confirm this.

The *Glow* model [11] uses invertible 1×1 convolutions in between coupling layer. They compare the performance of affine and additive transformations and find that affine coupling layers approximate the inflated manifold slightly better. However, the true distribution of the manifold is unknown, and thus the performance along it cannot be measured.

Nalisnick et al. [9] find that normalizing flows can assign higher likelihood to out-of-distribution data. They focus on affine coupling layers and can attributed this phenomenon to the scaling network. However, when they try additive coupling layers, the poor performance on out-of-distribution data persists. Whether additive or affine transformations perform better on detecting out-of-distribution data detection is not quantified.

2.5 Methods

In this section, we explain some methods used during the experiments.

2.5.1 Divergence Measures

If the true distribution p^* is known, we approximate the Kullback–Leibler (KL) divergence from the learned pdf q to p^* (see Definition 2.1). If the pdf is not known and we only have samples from both distributions (which is the case for the density of states), we compute the Wasserstein distance of first order, as it can be approximated from samples[1].

Definition 2.3. (Wasserstein distance of first order [49, Equation 10])
The Wasserstein distance of first order between two continuous distribution p^* and q is

$$D_{\text{Wasserstein}}(p^*,\, q) = \inf_{\pi \in \Gamma(p^*, q)} \int_{\mathbb{R}\times\mathbb{R}} |x-y| \, d\pi(x,y), \tag{2.16}$$

where $\Gamma(p^*, q)$ is the set of all distributions for which p^* and q are marginals.

2.5.2 Learned Density along the Manifold

We are interested in computing divergences for the ground truth density of the manifold and the learned density along the manifold. However, standard normalizing flows cannot give the density along the manifold. Since we know the ground truth of the manifold $\mathcal{M}$, we define q_M to be the learned manifold density with

$$q_M(m) = \begin{cases} \frac{1}{Z}\, q(m) & m \in \mathcal{M} \\ 0 & \text{else} \end{cases}, \tag{2.17}$$

$$\text{where } Z = \int_{\mathcal{M}} q(m)\, dm, \tag{2.18}$$

and $q(\cdot)$ is the learned density of the full space.

[1] We use the SciPy implementation. For details see: `https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wasserstein_distance.html`.

Chapter 3

Theoretical Analysis

In this chapter we introduce a notion to theoretically analyze the volume of states of a distribution and of VP and NVP normalizing flows. We start with simple cases to get an intuition and to introduce necessary tools. We then analyze normalizing flows and different manifolds and make hypotheses about the ability of normalizing flows to approximate manifolds with different properties.

3.1 Density of States

The *Density of States* is a concept used in physics to describe the volume of states for a given energy in a system [29]. We introduce it to the field of deep generative modeling, where it describes the distribution of log density.

Definition 3.1. (Density of States) Let p be a distribution. Then the Density of States (DoS) is the distribution of log density $\log p(A)$, where A is drawn from the density itself, i.e., $A \sim p$.

We will refer to the DoS of a distribution G as $p_{\log p(G)}$. The choice of base for the logarithm relates to the scaling of $\log p(G)$ and can thus be chosen arbitrarily. We use the natural logarithm. To get an intuition of the notion we show the DoS of common distributions and derive basic properties of the DoS. We also define certain terms that will be used in this section.

The DoS of a uniform distribution $\mathcal{U}[a, b]$ has its mass at exactly one density level, i.e., $\frac{1}{b-a}$. Thus, we can derive the following lemma which we show in Appendix A.1.5.

Lemma 3.1. (DoS of Uniform Distributions) The DoS of a uniform distribution $\mathcal{U}[a, b]$ has density $\delta(x + \log(b - a))$, where δ is the Dirac delta function.

A Gaussian distribution has different levels in its pdf. We show its form in Appendix A.1.2 and note the following lemma.

Lemma 3.2. (DoS of a Gaussian Distribution) The DoS of a normal distribution $\mathcal{N}(\mu, \sigma^2)$ has the (normalized) pdf

$$p_{\log p(\mathcal{N}(\mu,\sigma^2))}(c) = 2 \cdot p_{\chi^2}(-2c - \log(2\pi\sigma^2)), \tag{3.1}$$

where χ^2 has one degree of freedom.

From the pdf of the DoS of a normal distribution we can observe that the DoS is independent of the location μ. This holds for all distributions which we show in Appendix A.1.1. We plot the DoS of different Gaussians in Figure 3.1.

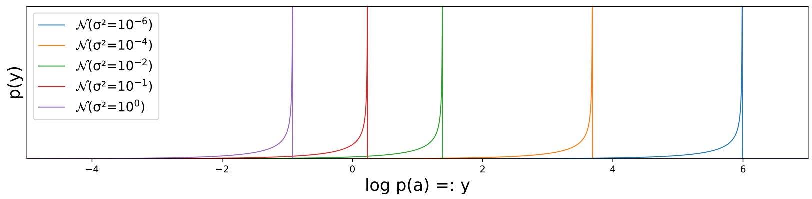

Figure 3.1: DoS's of normal distributions with different variances.

Definition 3.2. (Translation) A function $f : \mathcal{A} \to \mathcal{A}$ is said to be translated by $t \in \mathcal{A}$ when its graph is shifted by t without a change in shape, i.e. $f'(x) = f(x - t)$. A distribution p_A is said to be translated by t when its pdf is translated by t, respectively $A \sim p_A$ becomes $A + t$.

From the pdf of the DoS of normal distributions (Equation 3.1), we can see that the shape of the pdf is given by $2 \cdot p_{\chi^2}(-2c)$, as a function over c, and that a change in variances causes a translation to the pdf by $\log(2\pi\sigma^2)$. This property will be of interest when considering the DoS of inflated high-dimensional distributions, since under certain required conditions the DoS of a distribution inflated with Gaussian noise will preserve its shape regardless of the variance of the Gaussian.

Since the form of the DoS of a uniform and a Gaussian distribution is fundamentally different, we introduce the following definition to speak about it more easily.

Definition 3.3. (Degenerate-DoS / non-degenerate-DoS) A distribution p_A is called a *degenerate-DoS distribution*, if its DoS has a degenerated distribution, otherwise it is called a *non-degenerate-DoS distribution*. A continuous distribution p_A is called degenerated if it has its entire mass at a single point k, i.e. $P[A = k] = 1$ and $= 0$ elsewhere for $A \sim p_A$ [50, Definition 5.2.1].

3.1.1 Sampling the Density of States

Since there is not always a closed form of the DoS, we will do most of the computations with samples of the DoS. Since the DoS is a deterministic function $\log p(\cdot)$ applied to a random variable $A \sim p$, we can obtain a sample of the DoS from p by taking a sample a of A and applying $\log p(a)$.

3.2 Density of States of Simple High-Dimensional Distributions

In this section we derive properties of the DoS of simple high-dimensional distributions of i.i.d. random variables.

Let $X = (X_1, ..., X_D)$ be a D-dimensional random vector of i.i.d. variables X_i. The DoS of X can be written as

$$\log p(X) = \log(p(X_1) \cdot ... \cdot p(X_D)) = \sum_{i=1}^{D} \log p(X_i). \tag{3.2}$$

Since this is the sum of independent random variables, its pdf can be computed by convolution [51]

$$p_{\log p(X)}(a) = (p_{\log p(X_1)} * ... * p_{\log p(X_D)})(a), \tag{3.3}$$

where $*$ is the convolution, which is defined as:

Definition 3.4. (Convolution) The convolution $*$ of two functions f, $g : \mathbb{R} \to \mathbb{R}$ is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(s)\, g(x - s)\, ds. \tag{3.4}$$

Convolution is translation equivariant, which means that a translation by t of f or g results in a translation by t of $(f * g)$, more formally

Property 3.1. (Translation Equivariance of the Convolution) The convolution of two functions f and g is *translation equivariant*, that is it commutes with translation τ_t such that $f * (\tau_t g) = \tau_t(f * g) = (\tau_t f) * g$, where $(\tau_t f)(x) = f(x - t)$.

A proof is given in Appendix A.1.3. Using this property we can derive that the DoS of a D-dimensional normal distribution with covariance matrix $\sigma^2 I_D$ translates to the left with increasing σ while the shape is preserved, where I_D is the D-dimensional identity matrix, see Figure 3.2. This can be seen since the DoS of a one-dimensional Gaussian translates as a function of σ and since the DoS of a D-dimensional Gaussian is computed by convolution which is translation equivariant. A more detailed proof is given in Appendix A.1.4. This finding will become relevant when we analyze the DoS of a manifold that is inflated with noise that translates its DoS with different noise levels.

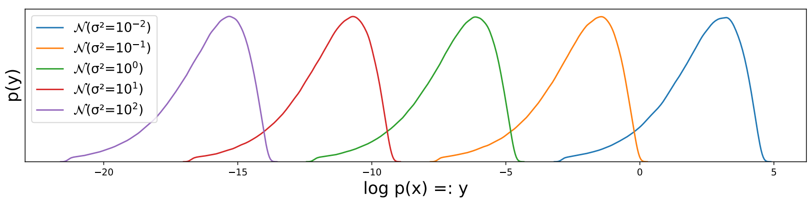

Figure 3.2: DoS's of 5-dimensional isotropic normal distributions $\mathcal{N}^5(\mu, \sigma^2 I_5)$ with different variances. Location μ has no effect.

Another observation from the sum that makes up the DoS of a high-dimensional distribution with i.i.d. random variables (Equation 3.2), is that the central limit theorem applies, which is

Definition 3.5. (Central Limit Theorem [52, Theorem 27.1]) Let $X^D = \{X_1, ..., X_D\}$ be a sequence of i.i.d. random variables with mean μ and finite positive variance σ^2. If $\overline{X} = \frac{1}{D}\sum_{i=1}^{D} X_i$, then

$$\lim_{D\to\infty} \sqrt{D}(\overline{X} - \mu) \sim \mathcal{N}(0, \sigma^2). \tag{3.5}$$

Thus we can state the following lemma.

Lemma 3.3. (DoS of a D-Dimensional Random Vector of i.i.d. Random Variables)
By the central limit theorem, the DoS $p_{\log p(X)}$ (Equation 3.2) has a limiting normal distribution $\mathcal{N}(D\mu, D\sigma^2)$ as $D \to \infty$ if $\sigma^2 > 0$, where μ and σ^2 are the mean and variance of $\log p(X_i)$.

We later show that an extended version of the central limit theorem can be applied to an inflated manifold with a fixed dimensionality, such that its DoS also becomes a limiting normal distribution.

The DoS of a uniform distribution is a Dirac distribution, which has zero variance. Thus the the central limit theorem does not apply since the variance needs to be positive. In fact the pdf of the DoS of a D-dimensional uniform distribution $\mathcal{U}^D[-a, a], a \in \mathbb{R}^+$ is $\delta(x + D\log(2a))$, which we show in Appendix A.1.5.

3.3 Density of States of Normalizing Flows

We continue with the analysis of the DoS of normalizing flows. We divide them into two categories and show the limits of one of them. Recall Formula 2.2 that governs normalizing flows

$$p(x) = p_Z(f(x)) \left|\det \frac{\partial f(x)}{\partial x}\right| = p_Z(f(x))\, v_f(x).$$

Lemma 3.4. (DoS of a Normalizing Flow) Let q_X be the distribution of a normalizing flow with transformation f. For $Z \sim p_Z$, it holds that $f^{-1}(Z) = X \sim q_X$. Thus, the DoS of a normalizing flow is the distribution of the following random variable

$$\log q_X(X) = \log p_Z(f(X)) + \log v_f(X) = \log p_Z(Z) + \log v_f(X). \tag{3.6}$$

From this Lemma we can infer that a normalizing flow has the DoS of its prior p_Z plus the logarithm of the volume element v_f. Since the DoS of the prior is fixed, the volume element is responsible for the change in DoS. Since v_f occurs in two different forms, we can make the following definitions.

Definition 3.6. (Non-Volume-Preserving Normalizing Flow) A normalizing flow for which $v_f(\cdot)$ is an arbitrary positive function is called *non-volume preserving*[1] (NVP) normalizing flow.

Definition 3.7. (Volume-Preserving Normalizing Flow) A normalizing flow for which $v_f(\cdot)$ is a constant positive function is called a *volume-preserving* (VP) normalizing flow, where the constant value of v_f results from the scale layer (see Section 2.2.2).

From these definitions, we can directly see that:

Corollary 3.1. (DoS of NVP normalizing flow) NVP normalizing flows can change the DoS of their latent distribution using the volume element v_f.

[1]The terminology is adapted from [2]. One might also call them *non-DoS-preserving*.

Corollary 3.2. (DoS of VP normalizing flow) The DoS of a VP normalizing flow has the shape of its latent distributions DoS up to the constant translation $\log v_f(\cdot)$.

The latter corollary describes the main limitation of VP normalizing flows, given that its transformation has maximum expressivity[1]. However, a VP normalizing flow can change the DoS for the pdf of a subset of the data space (e.g. the manifold), since it can use any volume-equivalent subset of the latent space. This can be best described with an example, see Figure 3.3. Note that when the dimensionality of the subset is small and the dimensionality of the latent space is large, a VP normalizing flow can utilize more latent dimensions and thus change the DoS of the subset more intensely (imagine more dimensions in Figure 3.3 and the pdf along a thread of fixed length, that is arranged arbitrarily in $\mathcal{Z}$.).

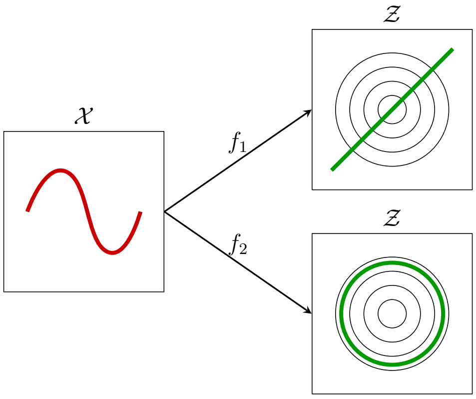

Figure 3.3: Sketch of how a VP normalizing flow can produce different DoS's for the same subset of the data space. The one-dimensional red subset (e.g. the manifold) in the data space $\mathcal{X}$ is mapped to the one-dimensional green subset in the latent space $\mathcal{Z}$ by VP transformations f_1 and f_2. The latent space is measured with the pdf of an isotropic two-dimensional Gaussian (black circles). The subset of f_1 intersects the Gaussian, resulting in a one-dimensional Gaussian for the red subset. The subset of f_2 lies on a level set of the Gaussian, resulting in a uniform distribution for the red subset.

From Lemma 3.3 we know that the DoS of a D-dimensional distribution becomes a limiting normal distribution when it consists of i.i.d. distributions for high D. Since this is fulfilled by the prior of a VP normalizing flow, we can make the following statement.

Corollary 3.3. (DoS of VP normalizing flows becomes a Limiting Normal Distribution) he DoS of a VP normalizing flow with a non-degenerate-DoS prior becomes a limiting normal distribution as $D \to \infty$ with a variance determined by the type of the latent distribution (see Lemma 3.3).

We will later see that under certain conditions, the DoS of a VP normalizing flow and of an inflated manifold have the same limiting normal distribution when the distribution of the (non-degenerate-DoS) prior of the VP normalizing flow is used for inflation.

[1] I.e., assuming that the VP normalizing flows are universal VP diffeomorphism approximators.

3.4 Density of States of Inflated Manifolds

We now analyze the DoS of inflated manifolds and build relations to the DoS of normalizing flows. We start by introducing an inflation method where the added noise does not interfere with itself or the manifold. A mathematical framework for this is given by Horvat and Pfister [23], which we will briefly summarize.

Let $\mathcal{M}$ be a d-dimensional manifold with distribution p_M embedded in a D-dimensional space such that it is not too entangled and not too curved. We add noise $E \in \mathbb{R}^{D-d}$ embedded in $\mathbb{R}^D$ in the normal space of $\mathcal{M}$, such that the new random variable becomes $X = M + E$, $M \sim p_M$. Generally, adding noise in the normal space of a manifold is empirically shown to be easy when $\frac{d}{D}$ is small [23, Appendix B.2].

Definition 3.8. (Q-normally reachable [23, Definition 4]) Let $\mathcal{M}$ be a d-dimensional manifold and N_m the normal space of $m \in \mathcal{M}$. Let $q_N(\cdot|m)$ be a density defined on N_m and denote by $N_{q_N(\cdot|m)}$ the domain of $q_N(\cdot|m)$. Denote the collection of all such densities as $Q := \{q_N(\cdot|m)\}_{m\in\mathcal{M}}$. We say $\mathcal{M}$ is Q-normally reachable if for all $x \in N_m$, x is almost surely determined by $m \in \mathcal{M}$.

Note that we use a simplified version of the original definition for Q-normally reachable. To familiarize with this concept, we refer to [23, Example 2].
We choose $\mathcal{M}$ such it is Q-normally reachable for noise E. For a realization of E and M, say e and m, $x = m + e$ is a realization of X with $x \in N_x$. By Theorem 5 of [23], the pdf evaluated at x can be expressed as

$$q_N(x) = q_N(x|m)\, p_M(m) = q_N(m + e|m)\, p_M(m). \tag{3.7}$$

Since m uniquely determines x, $q_N(m + e|m)$ simplifies to $p_E(e)$, and thus

$$q_N(x) = q_N(m + e) = p_E(e)\, p_M(m). \tag{3.8}$$

Note that $p_E(e)$ refers to the $\mathbb{R}^{D-d}$ dimensional pdf. Using this pdf, we can restate the random variable $\log q_N(X)$ that is distributed according to the DoS of the inflated manifold as

$$\log q_N(X) = \log p(E) + \log p_M(M). \tag{3.9}$$

We can now investigate the effect of the DoS of the noise and the DoS of the manifold separately. We start with a property which we need for some observations.

Property 3.2. (Identity of the Convolution [53]) The Dirac delta function δ is the identity function to the convolution. That is, $f * \delta = f$ for a function $f : \mathbb{R} \to \mathbb{R}$.

If E has an degenerate-DoS distribution, its DoS is described by a Dirac delta function. Since the pdf of $\log q_N(X)$ can be computed by convolution and since the Dirac delta function is the identity function for convolution, we can state the following lemma.

Lemma 3.5. (Preserving the DoS of the Manifolds under Degenerate-DoS Noise) If the manifold is Q-normally reachable for noise E and E has an degenerate-DoS distribution, the DoS of the inflated manifold preserves the shape of the DoS of the manifold.

An example of this can be seen in Figure 3.4. This can be explained intuitively by imagining the pdf of one-dimensional random variable M embedded in two dimensions, where we fill the empty dimension with uniform noise U such that the new random variable becomes (M, U). The pdf of (M, U) has the shape of the pdf of M, but is spread along the support of U. Therefore the pdf of M still accounts for the diversity in the DoS and thus the shape of the DoS does not change. The translation of the DoS occurs because of a different normalization.

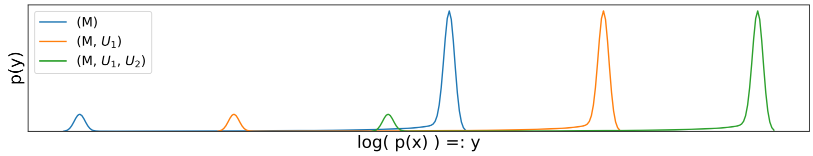

Figure 3.4: DoS of random variables (M), (M, U_1), (M, U_1, U_2), where M is of some distribution (e.g. the manifold distribution) and U_1, U_2 is i.i.d. uniform noise. The shape of the DoS does not change with dimensions containing only uniform respectively degenerate-DoS noise.

Since manifolds have an arbitrarily shaped DoS and the DoS of VP normalizing flows has a fixed shape, we hypothesize that noise with an degenerate-DoS distribution should not be used for inflation when using VP normalizing flows.

In Section 3.2, we have seen that some distributions (e.g. uniform and normal) translate their DoS's with different noise levels and preserve their shapes. Therefore, we can state the following theorem about the DoS of inflated manifolds.

Theorem 3.1. (Different Noise Levels Translate the Inflated Manifold DoS) If the manifold is Q-normally reachable for noise E, and E is chosen such that of the DoS of E only translates with different noise levels (e.g. normal noise, see Figure 3.2), then the DoS of the inflated manifold translates with different noise levels and preserves its shape.

Which follows since the pdf of $p_{\log q_N(X)}$ (see Equation 3.9) can be computed by convolution (see Equation 3.3) and that the convolution is translation equivariant (Property 3.1). A detailed proof is given in Appendix A.1.6. An example of this theorem can be seen in Figure 3.5. This finding is surprising since we had expected the effect on the DoS from inflation noise to the inflated manifold to vanish when the noise strength becomes sufficiently small. More precisely, we had expected the shape of the DoS of the inflated manifold to smoothly approach the DoS of the non-inflated manifold as the noise strength approaches zero[1].

This implies that it is not necessary to choose a tiny noise strength, as is often done in the literature (e.g. with a variance of approximately 10^{-6} for [1, 21, 34]), except to satisfy the requirements of Q-normal reachability. Our goal is not to evaluate the effect of different noise levels, but we will see in our experiments that as the noise level increases, the used normalizing flows perform better on approximating the manifold, the inflated manifold and their DoS's.

[1] In the limit $(\sigma \to 0)$, X is the non-inflated d-dimensional manifold in a D-dimensional space and thus the distribution of X is degenerated. The DoS of a degenerated distribution is "at infinity" which is not meaningful to define (consider $\log p_A(A)$, $A \sim p_A$, where p_A is degenerated).

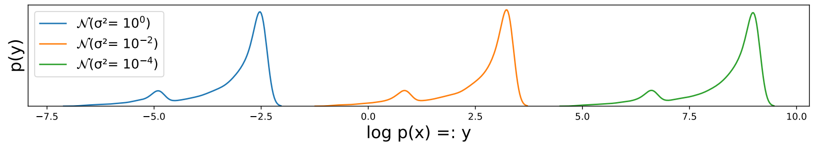

Figure 3.5: DoS's of 2-dimensional distributions (M, N), where M follows some distribution (e.g. the distribution of the manifold), and $N \sim \mathcal{N}(0, \sigma^2)$ is the noise used for inflation. It can be observed that different noise levels do not change the shape of the DoS, but translate it. The shape of the DoS results from the convolution of DoS of the Gaussian and the manifold, see Figure 3.1 respectively (M) in Figure 3.4.

3.4.1 Effect of Dimensionality

In Section 3.2 we show that the DoS of the distribution of a random vector with i.i.d. variables approaches a limiting normal distribution if the dimensionality of the random vector is large. In this subsection, we further decompose the DoS of the manifold and show that the DoS of an inflated manifold approaches a limiting normal distribution under certain conditions. We start with a definition of the Gram matrix and a general assumption about the problem setting of deep generative models.

Definition 3.9. (Gram Matrix) Let $g : A \to B$ for some sets A and B, and let $J_g(m)$ be the Jacobi matrix of g at position m. We denote the Gram matrix of $g(m)$ as

$$G_g(m) := J_g(m)^T J_g(m) \,. \tag{3.10}$$

In affiliation with the problem setting of deep generative models of Horvat and Pfister [27, Section 2], we assume that the manifold is generated by a transformation $g^{-1}(L) = M$, where $L \sim \mathcal{N}(0, I_d)$ and that g^{-1} is invertible, meaning that g can decompose M into independent random variables of standard normal distribution such that $p(g(m)) \overset{\dagger}{=} |\det G_g(m)|^{\frac{1}{2}} \prod_{i=1}^{d} p_{\mathcal{N}}(g_i(m))$, where $p_{\mathcal{N}}$ is the standard normal distribution and $m \in \mathcal{M}$. Using g, the DoS of the inflated manifold can be further decomposed

$$\log q_n(X) = \log p(E) + \log p(M) = \sum_{i=1}^{D-d} \log p(E_i) + \sum_{i=1}^{d} \log p_{\mathcal{N}}(g_i(M)) + \underbrace{\frac{1}{2} \log |\det G_g(M)|}_{G_M} \,. \tag{3.11}$$

With this decomposition we arrive at our main theorem.

Theorem 3.2. (DoS of Inflated Manifolds) If the manifold is Q-normally reachable for a non-degenerate-DoS noise E, then for a fixed manifold dimensionality d and for $\mathbb{E}[\,|\,G_M - \mathbb{E}[G_M]\,|^3\,] < \infty$, the DoS of an inflated manifold approaches a limiting normal distribution with the variance of the DoS of the inflation noise when $D \to \infty$, respectively $\frac{d}{D} \to 0$.

A proof is given in Appendix A.1.7 and a visualization for it can be seen in Figure 3.6. The condition $\mathbb{E}[\,|\,G_M - \mathbb{E}[G_M]\,|^3\,] < \infty$ is a restriction on the embedding function g^{-1} of the manifold and can be interpreted as: there should be no areas of significant probability in the latent distribution of the manifold that are compressed to a point or stretched to an area of infinite space on the manifold.

†By change of variables and independence of the elements of $g(\mathcal{M})$.

This theorem means that when the dimensionality d of the manifold is small and the dimensionality D of the embedding space is large, as is often the case for natural data (e.g. $\frac{d}{D}$ is estimated to be $\frac{13}{784}$ for MNIST and $\frac{43}{150{,}528}$ for ImageNet, see [54]), the DoS of the inflated manifold has a limiting normal distribution when non-degenerate-DoS noise is used for inflation.

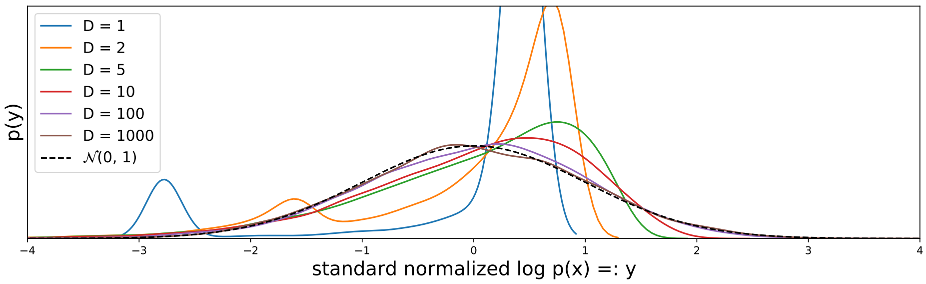

Figure 3.6: Sampled DoS's of an inflated manifold $X = (M, N_2, ..., N_D)$, where M is the random variable of a manifold, and $N_i \sim \mathcal{N}(0, 10^{-6})$. $D = 1$ means $X = (M)$. While the manifold dimensionality remains constant, the dimensionality of the data space and hence the dimensionality of the inflation noise increases. As the dimensionality increases, the DoS of X approaches a limiting normal distribution.

DoS matching of VP normalizing flows

The variance of the limiting normal distribution of the DoS of inflated manifolds depends on the type and parameters of the noise distribution, as is the case for the limiting normal distribution of the DoS of VP normalizing flows. Thus, if inflation noise is used that matches the prior of the normalizing flow, the DoS of the manifold and the DoS of the VP normalizing flow have a limiting normal distribution with the same variance[1]. The location of the target DoS can be matched by VP normalizing flows, since they can arbitrarily translate their DoS (see Corollary 3.2).

We thus hypothesize, that when $\frac{d}{D}$ is small, VP and NVP normalizing flows can perform comparably, since the DoS of the inflated manifold can be matched by VP normalizing flows. In contrast, when $\frac{d}{D}$ is large, NVP normalizing flows should outperform VP ones because the target DoS and the DoS of the VP normalizing flow are too different.

[1]The variance of the limiting normal distribution of the DoS of a multivariate Gaussian $\mathcal{N}^D$ is independent of the variance of $\mathcal{N}^D$, see Figure 3.2. Thus, if a Gaussian distribution is used for inflation and as latent distribution, the parameters do not influence the variance of the limiting normal distribution.

3.5 Hypotheses

In this section we summarize the hypotheses made during the theoretical analysis.

Hypothesis 3.1. If the manifold is Q-normally reachable for noise E then when $\frac{d}{D}$ is large, NVP normalizing flows should outperform VP normalizing flows in estimating the inflated distribution.

This follows from the restriction on the DoS of VP normalizing flows (see Corollary 3.2).

Hypothesis 3.2. If the manifold is Q-normally reachable for noise E and E matches the distribution of the prior of the normalizing flow, then when $\frac{d}{D}$ is small, VP and NVP normalizing flows can perform comparably in estimating the inflated distribution.

This hypothesis follows since the limitation of VP normalizing flows (see Corollary 3.2) is lessened by Theorem 3.2.

Hypothesis 3.3. If the manifold is Q-normally reachable for noise E, then E should not be degenerate-DoS noise when VP normalizing flows are used.

Which we hypothesis as VP normalizing flows cannot change the shape of the DoS (Corollary 3.2) and that degenerate-DoS noise preserves the shape of the DoS of the manifold (Lemma 3.5).

Chapter 4

Empirical Analysis

4.1 Setup

We conduct experiments on artificial data. We embed a d-dimensional manifold of ground truth distribution $p_\Pi : \mathbb{R}^d \to \mathbb{R}^+$ in a D-dimensional space and then inflate it with noise. Let $p_X : \mathbb{R}^D \to \mathbb{R}^+$ be the resulting density, and let $\psi : \mathbb{R}^d \to \mathbb{R}^D$ be the embedding function such that $p_M(x) = p_\Pi(\psi^{-1}(x))\, v_{\psi^{-1}}(x)$ denotes the d-dimensional density in the D-dimensional space, where $v_{\psi^{-1}}$ is the volume element of ψ^{-1}. We are interested in learning p_X and p_M from samples of p_X. Unless mentioned otherwise, we use normal noise for inflation, which we add in the full ambient space of the samples. Since this smooths the manifold, we assume that there is a perfect deflation method that reverses the smoothing. Finding this method is ongoing research (cf. Section 2.3), we thus consider the smoothed manifold as the target manifold. We apply different inflation noise levels to the same ground truth manifolds, leading to target manifolds of differently smoothed distributions and thus different DoS's of the manifold (DoS's shown in Appendix A.1.10). Since we hypothesize that VP and NVP normalizing flows differ in their ability to approximate a distribution well because NVP normalizing flows can adjust the DoS and VP normalizing flows cannot, using different target DoS's means we can test our hypotheses against a greater variety of DoS's.

For the normal noise we use $\sigma^2 = 0, 10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}$. For the no-noise case ($\sigma^2 = 0$) the KL divergence from the target distribution to the inflated distribution and the Wasserstein distance of the respective DoS's do not exist[1]. The embedding spaces have dimensionality $D = 2, 36, 100, 196$. In all our experiments, we fix $d = 1$ and thus $\frac{d}{D}$ is large for $D = 2$ and small for $D = 36, 100, 196$. For VP normalizing flow we use NICE [1] and RealVP, and for the NVP normalizing flow we use RealNVP [2]. A detailed description of the model and training can be found in Appendix A.3.

[1]Without inflation noise, the target distribution is of dimensionality $d\,(= 1)$ and the learned distribution of dimensionality $D\,(> 1)$. However, the KL divergence is defined on the same domain, which is not fulfilled. The DoS for the target distribution is not meaningful to define, since it "resides at infinity", see Footnote [1].

4.1.1 Artificial Data Generation

We use three basic one-dimensional manifolds distribution p_Π: a uniform distribution ($\mathcal{U}[-0.5, 0.5]$), a *von Mises* distribution [55] ($\kappa = 0.5, \mu = 0$), and a *Smooth Step* distribution, which we defined ourselves, see Appendix A.4. The pdfs of the target manifolds can be seen in Figure 4.1 and their DoS's can be seen in Appendix A.5. The Smooth Step distribution is named after the shape of its pdf and is used to have a distribution of complex DoS, i.e. it consists of two delta function terms and a smooth connection between them. The delta function terms in the DoS disappear when noise is added, as shown in Appendix A.1.9.

We embed $\Pi \sim p_\Pi$ in $\mathbb{R}^D$ by rotating it with rotation matrix $R_D \in \mathrm{SO}(D)$ like

$$\psi(\Pi) = R_D \cdot (\Pi, \underbrace{0, ..., 0}_{D-1})^T, \tag{4.1}$$

where R_D is fixed for a given D[1]. Note that rotation matrices have a determinant of 1 [56, Definitions], thus the volume element of ψ is 1 and $p_M(x) = p_\Pi(\psi^{-1}(x))$.

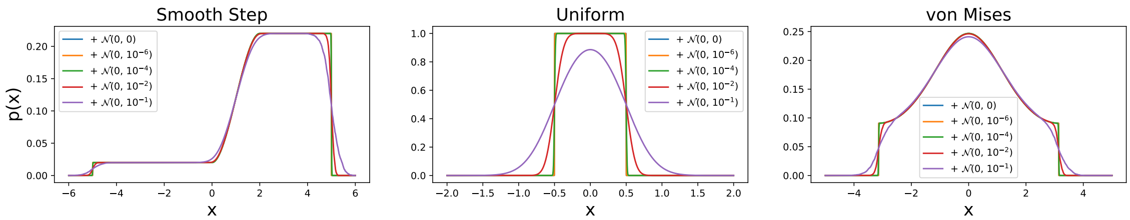

Figure 4.1: Target manifolds with different amount of noise added to them.

[1]For implementation details see `https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.special_ortho_group.html`.

4.2 Goodness of Fit

In this section, we test our hypotheses and measure further performance differences between VP and NVP normalizing flows, which we briefly summarize.

In Section 4.2.1 we compare the KL divergence of the learned distribution q_X to the inflated manifold p_X and the Wasserstein distance of their respective DoS's, where q_X is trained on samples of p_X. The manifold is inflated with normal noise, thus the inflation noise matches the distribution of the prior of the normalizing flows used. We then compare the distribution of q_X along the ground truth manifold to the manifold distribution p_M and the Wasserstein distance of their respective DoS's in Section 4.2.2.

In the general learning problem, we can measure how well a normalizing flow approximates an inflated distribution, but cannot measure how well the distribution along the manifold is approximated. Since we know the true topology and distribution of the manifold, we can measure it and thus quantify whether a good approximation of the inflated manifold predicts a good approximation of the manifold distribution in Section 4.2.3.

Finally, we test how well the models approximate a distribution when degenerate-DoS noise is used for inflation in Section 4.2.4. This section ends with a summary and further discussion in Section 4.2.5.

Throughout this section, we speak of *VP models* when we mean NICE and RealVP, and of *NVP model* when we mean RealNVP.

4.2.1 Performance on the Inflated Manifold

In this section, we evaluate the performance of our models on the inflated manifold in terms of KL divergences and Wasserstein distance between the respective DoS's.

In this section, we speak of *the performance of a model q_X for a distribution p_X*, meaning the KL divergence from q_X to p_X, where q_X is trained on samples of p_X. We speak of *the approximation of the DoS of a model q_X for a distribution p_X*, meaning the Wasserstein distance between the DoS's of q_X and of p_X.

A comparison between the performances of the models is shown in Figure 4.2. There is a general trend that performance improves as the noise level increases. This does not mean one should use large noise levels because this could smooth the manifold beyond recognition. However, our objective is not to find good noise levels, thus we will not discuss this in detail.

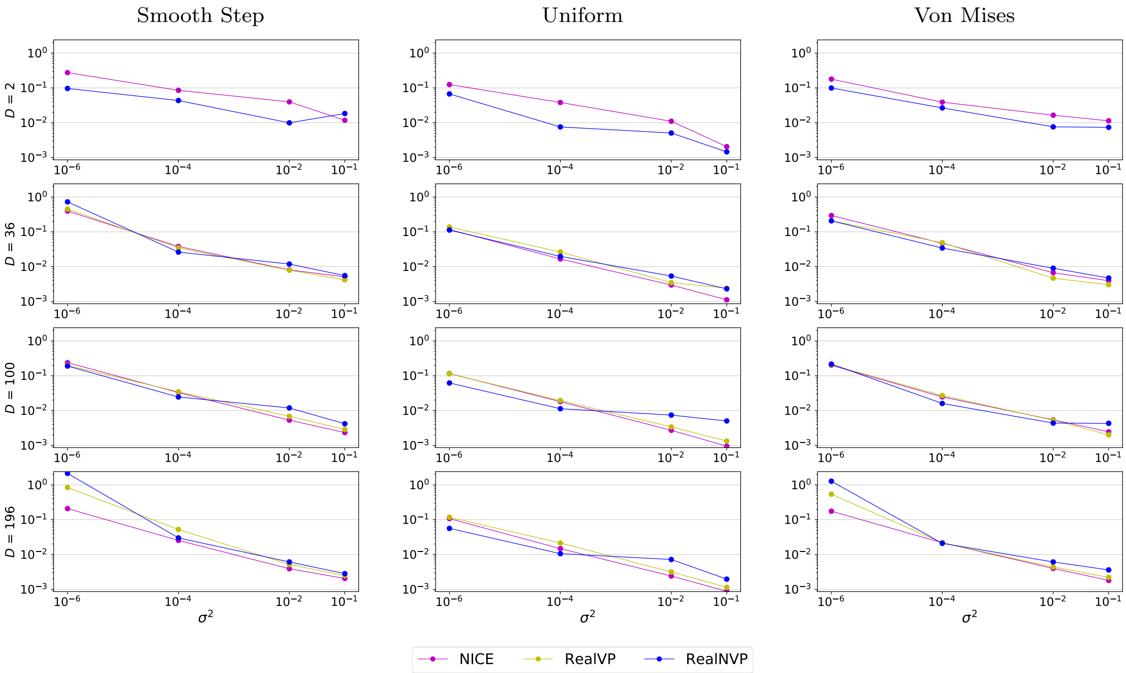

Figure 4.2: **KL divergences to the inflated distribution divided by D.** KL divergence from the learned distribution to the inflated distribution for all used manifolds (see subcaptions). Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale. For $D = 2$, RealVP coincides with NICE. For all divergences measured, see Appendix A.3.

We discuss the cases of large and small $\frac{d}{D}$ separately, since our hypotheses are split that way.

Large $\frac{d}{D}$

For large $\frac{d}{D}$ (i.e. $D = 2$), RealNVP almost always outperforms the VP models on the inflated manifold (first row $D = 2$, Figure 4.2). This confirms Hypothesis 3.1, which states that NVP normalizing flow will outperform VP normalizing flows for large $\frac{d}{D}$ because the difference in DoS is too large as shown in Section 3.4.1. This difference is quantitatively expressed in Figure 4.4 (first row, $D = 2$), where it can be seen that RealNVP matches the DoS, while the VP models do not. Qualitatively, the difference can be seen in the DoS plots in Figure 4.3 (first row, $D = 2$) where RealNVP makes adjustments to match the DoS, while the VP models have a fixed shape DoS that cannot match the target DoS. In particular, from the DoS of RealNVP (first row, column two and three, Figure 4.3), we observe that RealNVP adjusts its DoS to match the large bump on the right side and makes small adjustments for the small bump left. However, this does not happen in for all noise levels (e.g. first and fourth column) and for DoS's of other manifolds (e.g. first row in von Mises in Appendix A.4), which indicates an unstable behavior of the volume element of RealNVP which we will dig into in Section 4.3.

This results implies that using NVP normalizing flow when $\frac{d}{D}$ is large is beneficial.

Small $\frac{d}{D}$

For small $\frac{d}{D}$ (i.e. $D = 36, 100, 196$), the performances of VP and NVP normalizing flows are comparable (bottom rows $D = 36, 100, 196$, Figure 4.2). This confirms Hypothesis 3.2, which states that for small $\frac{d}{D}$, VP and NVP normalizing flows can perform equally well because the main limitation of VP models of having a fixed DoS is mitigated since the target DoS and the DoS of the VP models become are alike. The effect of easier DoS matching can be seen in Figure 4.3 (bottom three rows, $D = 36, 100, 196$), where we see the expected Gaussian-like target DoS with the same variance as the DoS of the VP models. From the Wasserstein distances between the learned and target DoS's in Figure 4.4 (bottom rows, $D = 36, 100, 196$), we also see that the VP models improve their DoS matching with smaller $\frac{d}{D}$ (cf. rows $D = 2$ to $D = 36, 100, 196$), but also with increasing inflation noise.

DoS of RealNVP also tends to match the DoS of the target distribution as noise levels increase, but with more fluctuations. From the DoS plots in Figure 4.3 (bottom rows, $D = 36, 100, 196$), we observe that RealNVP tend to translate its DoS away from the target DoS, leading to a worse approximation of the DoS. Since this translation of the DoS of RealNVP is unexpected and since we observe more such behavior in other experiments, we investigate RealNVPs behavior in more detail in Section 4.3, where we find that RealNVP behaves unstable due to a very steep loss space.

These results suggests that VP normalizing flows are slightly better than NVPs for small $\frac{d}{D}$, which will become more evident in the next sections.

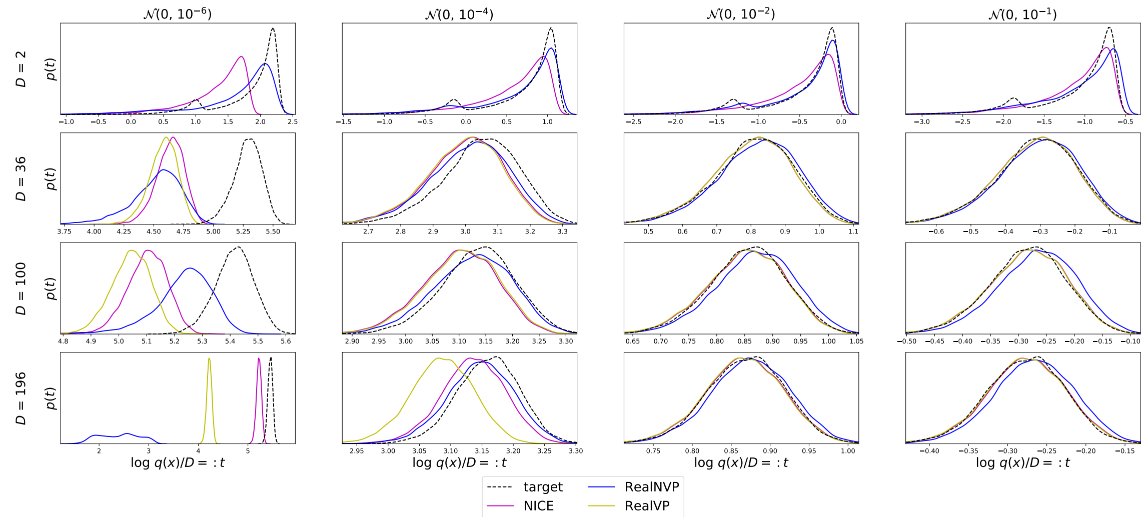

Figure 4.3: **Target and learned DoS's** for the Smooth Step distributions. For $D = 2$, RealVP coincides with NICE. The DoS of the target distribution without noise is undefined. For $D = 2$, RealNVP slightly adjusts its DoS's to the target DoS's (see two dents at $\mathcal{N}(0, 10^{-4})$ or $\mathcal{N}(0, 10^{-2})$). For small $\frac{d}{D}$ ($D = 36, 100, 196$), the DoS's of the VP models and the target DoS's are more similar and therefore easier to match. Still, with higher noise, RealNVP often translates its DoS away from the target DoS. In general, the DoS is matched worse with smaller noise. Plots for the other manifolds are shown in Appendix A.4

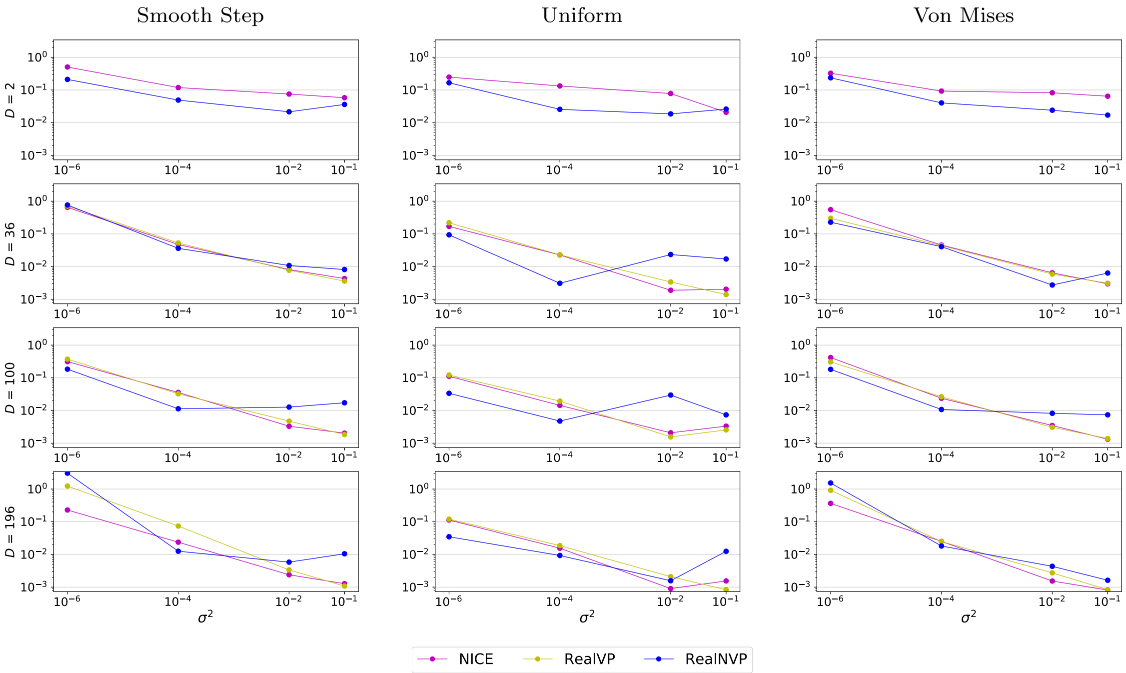

Figure 4.4: **Wasserstein distance between the DoS's.** Wasserstein distance of the DoS of the learned distribution and the DoS of the inflated distribution divided by the dimensionality for all used manifolds (see subcaptions). Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale. For $D = 2$, RealVP coincides with NICE. For all divergences measured, see Appendix A.3.

4.2.2 Performance on the Manifold

The main task for normalizing flows is to learn the density and the topology and the density of the manifold. In this section we study how well the models approximated the density along the ground truth manifold. Standard normalizing flows cannot give the density along the manifold p_M, therefore we use the method described in Section 2.5.2 to obtain q_M, which is the normalized density of q along the manifold[1], when q is trained on samples of the inflated manifold. We also compute the Wasserstein distance between the DoS of q_M and p_M. To obtain samples from q_M for the Wasserstein distance we use inversion sampling on a discretized version of q_M as described in Appendix A.1.8.

We speak of *the performance of a model q along the manifold*, meaning the KL divergence from q_M to p_M. We speak of *the approximation of the DoS of the manifold from a model*, meaning the the Wasserstein distance between the DoS of q_M and p_M. We present a quantitative comparison for the performances of the models along the manifold in Figure 4.5 and for the approximation of the DoS along the manifold in Figure 4.7, and a qualitative comparison of learned pdfs and DoS's in Figure 4.6 and 4.8, respectively.

[1]I.e. $q_M(x) = 1/Z\, q(x)$ for $x \in \mathcal{M}$ and 0 elsewhere, where $1/Z$ is the normalization.

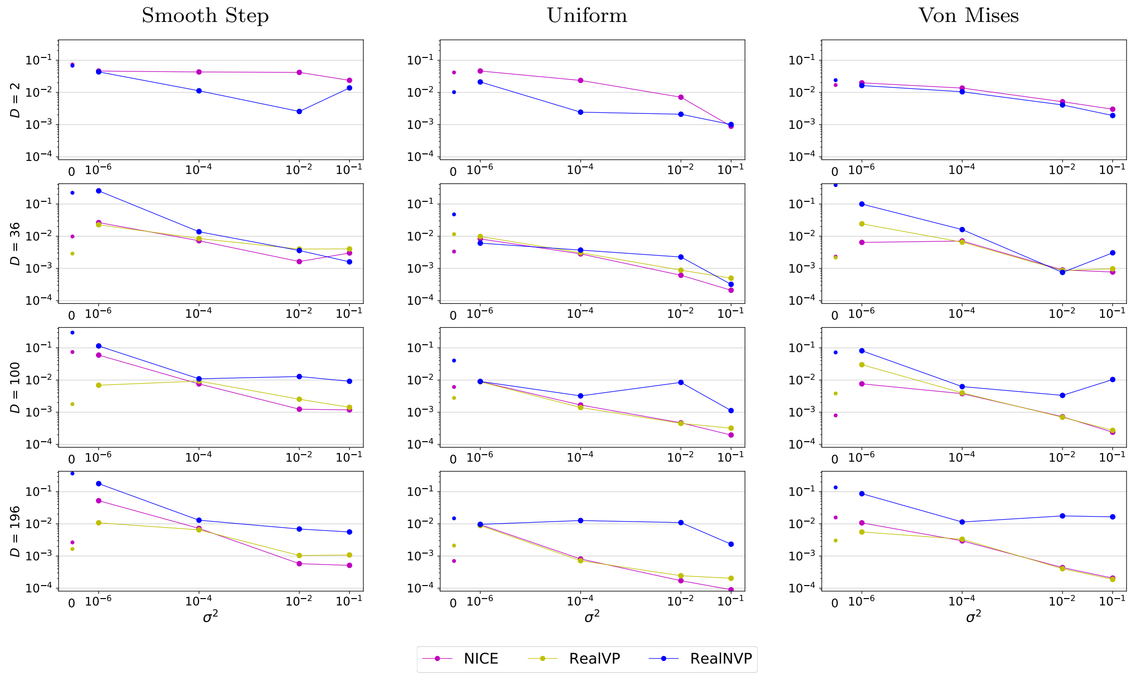

Figure 4.5: **KL divergences along the manifold.** KL divergences from the learned distribution along the manifold q_M to the distribution of the manifold for all used manifolds (see subcaptions). For further details on the computation of q_M see Section 2.5.2. Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale. For $D = 2$, RealVP coincides with NICE. For all divergences measured, see Appendix A.3.

We first note common phenomenons and then discuss large and small $\frac{d}{D}$ separately, since our hypotheses are split that way.

The performance of all models along the manifold increases as noise levels increase. Since we consider the smoothed, i.e. noisy, version of the manifold to be the target manifold, we assume that this happens because smooth manifolds are easier to be learned. This does not mean one should use large noise levels because this could smooth the manifold beyond recognition. However, noise levels are not an objective of our research, and thus we will not discuss this effect in detail.

The learned pdfs in Figure 4.6 show unexpected ups and downs for low noise levels (columns one to three) and smoother pdfs for high noise levels (columns four and five). Since the DoS shows the density of the log density, these ups and downs are also visible in the learned DoS along the manifold in Figure 4.8. The reason for this is explained in Section 4.3.1, where we find that for small noise (i.e. $\mathcal{N}(0,0)$, $\mathcal{N}(0,10^{-6})$, $\mathcal{N}(0,10^{-4})$) the models map the manifold to the latent space in a "zigzag" pattern leading to "zigzag" pdfs, and for larger noise (i.e. $\mathcal{N}(0,10^{-2})$, $\mathcal{N}(0,10^{-1})$) map the manifold in a straight way to the latent space leading to a smooth density (see Figure 4.18).

Large $\frac{d}{D}$

For large $\frac{d}{D}$ (i.e. $D = 2$), RealNVP almost always outperforms the VP models along the manifold (first row $D = 2$, Figure 4.5), which is consistent with the performance on the inflated manifold (first row, Figure 4.2). From the learned pdfs along the manifold in Figure 4.6 (first row $D = 2$), it is evident that RealNVP often matches the pdf of the manifold well, while the VP models do not. This is also evident in the plots of the DoS along the manifold in Figure 4.8 (first row $D = 2$), where RealNVP often produces a shape that resembles the DoS of the manifold (e.g. row $\mathcal{N}(0, \mathcal{N}(0, 10^{-4}))$ or $\mathcal{N}(0, \mathcal{N}(0, 10^{-2})))$, while NICE only matches the bump on the right and otherwise produces strange DoS (except for row $\mathcal{N}(0, \mathcal{N}(0, 10^{-1})))$.

Since the findings for the model performances for the inflated manifold (Section 4.2.1) are similar, these results imply that using NVP normalizing flow when $\frac{d}{D}$ is large is beneficial.

Small $\frac{d}{D}$

For small $\frac{d}{D}$ (i.e. $D = 36, 100, 196$), NICE and RealVP perform better along the manifold than RealNVP most of the time (bottom rows $D = 36, 100, 196$, Figure 4.5). From the learned pdfs along the manifold (bottom rows $D = 36, 100, 196$ in Figure 4.6), it is evident that the VP models perform better than RealNVP because RealNVP allocates high probabilities to unreasonable regions (e.g. bottom rows $D = 36, 100, 196$ and $\mathcal{N}(0, 10^{-6})$ or $\mathcal{N}(0, 10^{-1})$, Figure 4.6). This effect increases as $\frac{d}{D}$ decreases. The pdfs for the other distributions show the same pattern, see Appendix A.5. Since this is unexpected, we investigate it in Section 4.3, where we find that RealNVP behaves unstable due to a very steep loss space.

From the comparison of the approximation of the DoS of the manifolds (quantitative in Figure 4.7 and qualitative in Figure 4.8, bottom rows $D = 36, 100, 196$), we see that all models approximate the DoS of the manifold poorly, although the approximation of the VP models are slightly better than RealNVPs. In some cases, we observe that, unlike RealNVP, the VP models mimic the DoS of the manifold, though they do not actually match it (e.g. in Figure 4.8, second row, last column). This shows that VP models can adjust the DoS of a subset to match the DoS of a subset of the target distribution. An intuitive explanation for this is given in Figure 3.3, where we show that a subset of the data space mapped to the latent space can yield different pdfs depending on this mapping.

These findings imply that until the unstable behavior of RealNVP is fixed, the usage of VP normalizing flows is beneficial when $\frac{d}{D}$ is small.

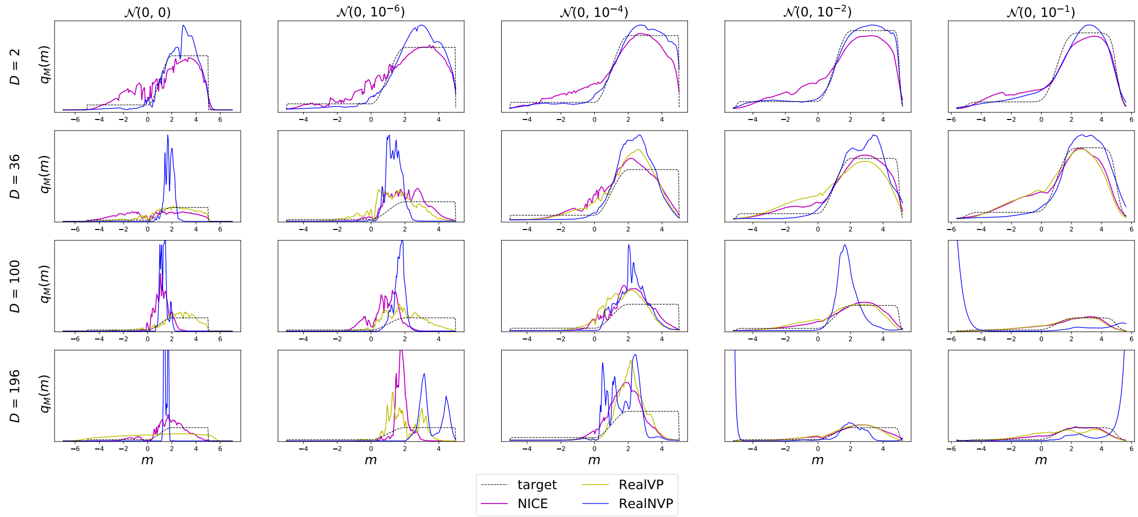

Figure 4.6: **Learned and target pdfs** of the Smooth Step distributions. m are the points along the manifold. q_M is the learned or true pdf along the manifold. For $D = 2$, RealVP coincides with NICE. For the learned pdfs along the manifold of all manifolds see Appendix A.5

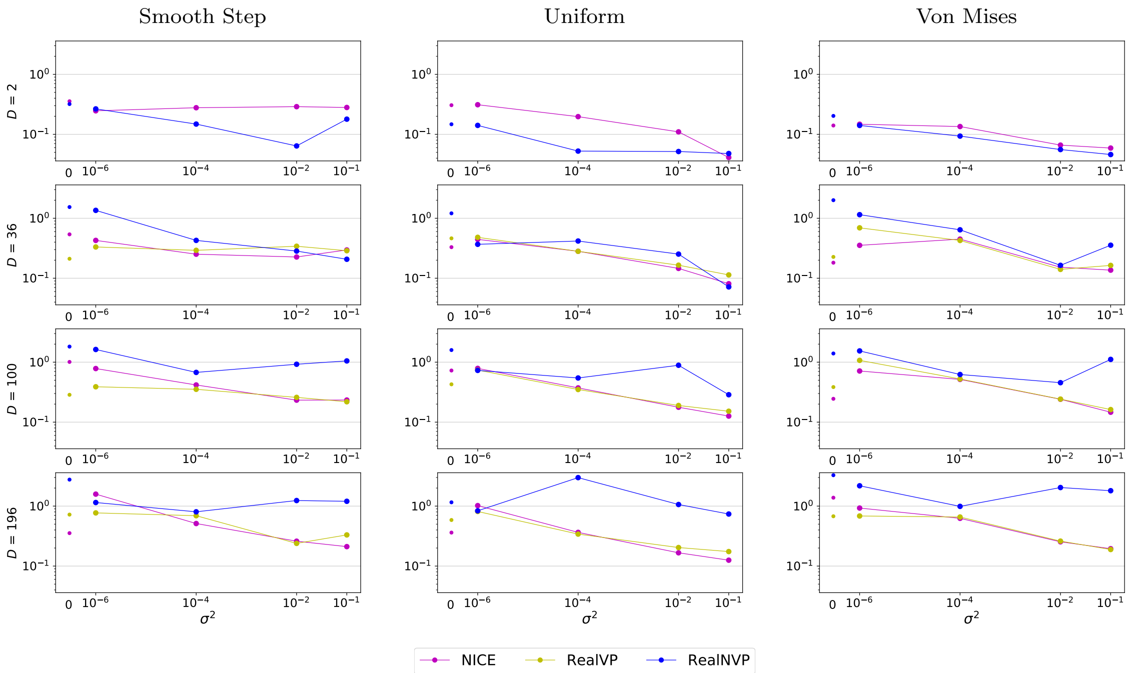

Figure 4.7: **Wasserstein distance between the DoS's along the manifold**. Wasserstein distance of the DoS's of the learned density along the manifold q_M and of the density of the manifold. For creation of the learned DoS along the manifold see Appendix A.1.8. Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale. For $D = 2$, RealVP coincides with NICE. For all divergences measured, see Appendix A.3.

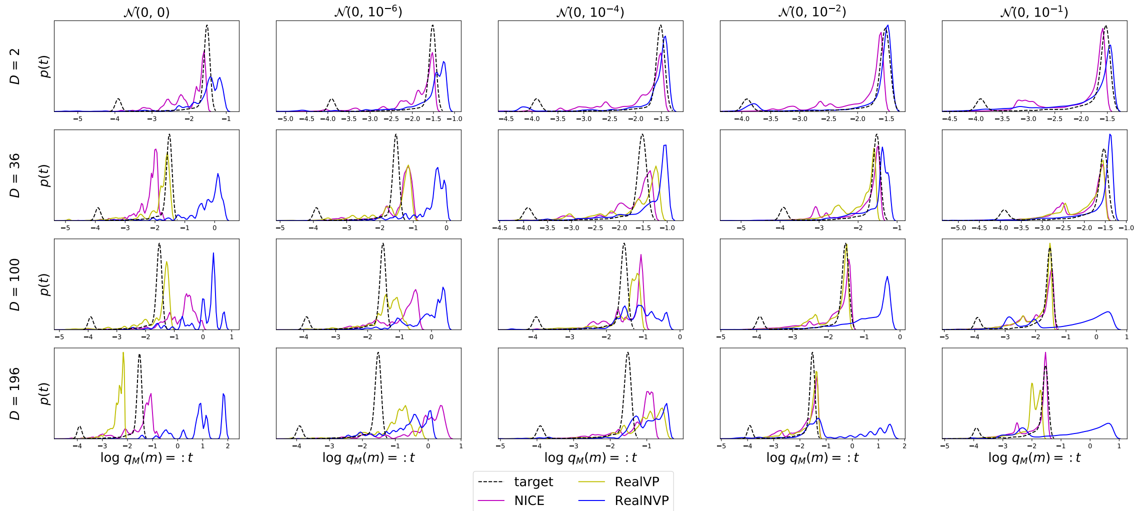

Figure 4.8: **Learned and target DoS along the manifold** of the Smooth Step distributions. m are the points along the manifold. $q_M(m)$ is the learned or true density along the manifold. For $D = 2$, RealVP coincides with NICE. For the learned DoS along the manifold of all manifolds see Appendix A.6

4.2.3 Does the Likelihood Predict the Performance on the Manifold?

The general problem setting for normalizing flows is to learn the density and the topology of the manifold well. However, we can only measure how well a normalizing flow approximates an inflated distribution, but not how well the density of the manifold is approximated. This is important for practical applications, if we are given in-distribution data and want to know their likelihood. Since we know the true topology and distribution of the manifold, we can measure how well the models approximate the density of the manifold. By comparing the KL divergence for the inflated and non-inflated manifold in Figure 4.2 and 4.5 (best seen in Appendix A.3 column "KL Div / D" and "KL Div on Manifold"), we notice that better performance on the inflated manifold does not necessarily imply better performance on the manifold. Since normalizing flows are often preferred over others if their likelihood for the inflated manifold is better [2, 6, 10, 11, 47], we quantify whether a better likelihood predicts a better approximation of the density of the manifold[1].

We express this prediction using the *false discovery rate* for a given model. That is, how often does a better performance of a model on the inflated manifold falsely predicts that this model performs better on the manifold than another model. The results are shown in Table 4.1.

Definition 4.1. (False Discovery Rate [57, Definition 2.1]) The false discovery rate (FDR) is the proportion

$$\frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FP}}, \tag{4.2}$$

where FP are false positives and TP are true positives, which in our case is

[1]The log-likelihood is the negative KL divergence minus the (constant) entropy of the target distribution (see Equation 2.1), thus it is proportional to the KL divergence from our plots.

$$\mathrm{TP}_\alpha = \sum_{\substack{p_M \in \mathbb{M} \\ \beta \in \mathbb{Q}\backslash\{\alpha\}}} \mathbb{1}_{[\mathrm{KL}(\alpha|p_{M+E}) \ < \ 0.9\,\mathrm{KL}(\beta|p_{M+E}) \wedge \mathrm{KL}(\alpha_\mathcal{M}|p_M) < \mathrm{KL}(\beta_\mathcal{M}|p_M)]}, \tag{4.3}$$

$$\mathrm{FP}_\alpha = \sum_{\substack{p_M \in \mathbb{M} \\ \beta \in \mathbb{Q}\backslash\{\alpha\}}} \mathbb{1}_{[\mathrm{KL}(\alpha|p_{M+E}) \ < \ 0.9\,\mathrm{KL}(\beta|p_{M+E}) \wedge \mathrm{KL}(\alpha_\mathcal{M}|p_M) > \mathrm{KL}(\beta_\mathcal{M}|p_M)]}, \tag{4.4}$$

where α is the model of interest, p_{M+E} is a manifold inflated with noise E, $\mathrm{KL}(\alpha|p_{M+E})$ is the KL divergence from α to p_{M+E}, when α is trained on samples from p_{M+E}, and $\alpha_\mathcal{M}$ is the density of the corresponding α along the manifold. The sets over which the sum iterates are $\mathbb{Q}$, which is the set of all models, and $\mathbb{M}$, which is the set of all manifold distributions, The factor 0.9 for $\mathrm{KL}(\beta|p_{M+E})$ exists to allow for some slack, i.e. model α need to perform at least 10% better than β, such that it should also have a better performance along the manifold. FP_α in words is: how often does a model α approximates an inflated manifold p_{M+E} better (first inequality) and approximates the (deflated) manifold p_M worse (second inequality), as compared to all other models on all manifolds and all inflation strengths.

Dimension	2	36	100	196	large $\frac{d}{D}$	small $\frac{d}{D}$
NICE	**100**	29	25	32	**100**	29
RealVP	-	60	27	08	-	33
RealNVP	09	**64**	**100**	**100**	09	**86**

Table 4.1: **False discovery rate** in percent for a model to perform better on the manifold given that it performs better on the inflated manifold as compared to the other models in pairwise comparisons. 0 means that the performance on the inflated manifold always correctly predicts the performance on the manifold, 50 means that this prediction is correct halve of the time, 100 means it is always wrong. We mark worst scores. For $D = 2$, RealVP coincides with NICE. The two columns on the far right summarize the results for the different dimensions, i.e. "large $\frac{d}{D}$" is for $D = 2$ and "small $\frac{d}{D}$" is for $D = 36, 100, 196$.

For large $\frac{d}{D}$ (i.e. $D = 2$), the performance of RealNVP on the inflated manifold correctly predicts the better performance on the manifold, while the performance of NICE does not (second column from right "large $\frac{d}{D}$", Table 4.1). For small $\frac{d}{D}$ ($D = 36, 100, 196$), the performance of RealNVP is usually expertly bad, which means that one should almost always reject RealNVP and choose another model, while the performance of the VP models on the inflated manifold often correctly predicts the performance on the manifold (rightmost column "small $\frac{d}{D}$", Table 4.1).

Since the ground truth of the manifold is usually unknown, this suggests to use NVP normalizing flows for large $\frac{d}{D}$ and VP normalizing flows for small $\frac{d}{D}$ when the task is to get a good approximation of the density of the manifold. It also suggests that the performance on the inflated manifold should generally not be used as a proxy to determine how well the distribution of a manifold is approximated. To close this gap, we should develop natural datasets with known manifold distribution such that we can directly measure the divergence to the distribution of interest, i.e. the (deflated) manifold.

4.2.4 Degenerate or Non-Degenerated-DoS Noise for Inflation

We now test Hypothesis 3.3, which states that when using VP normalizing flows, degenerate-DoS noise (e.g. uniformly distributed) should not be used, since the difference in DoS is too high as shown by Lemma 3.5. Instead noise that matches the distribution of the prior of the normalizing flow (usually normally distributed) should be used. We test this by training our models with uniform (i.e. degenerate-DoS) and normal (i.e. prior matching) noise for inflation. We choose the parameters of the noise such that the signal-to-noise ratios are equivalent (i.e. the noises have the same variance), which is $\mathcal{N}(0, \sigma^2)$ and $\mathcal{U}[-\sigma\sqrt{3}, \sigma\sqrt{3}]$, see Definition 4.2 and Figure 4.9. We measure different divergences for the trained models and the ground truth distribution and show the difference qualitatively in Figure 4.10.

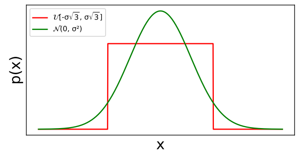

Figure 4.9: Pdf of a normal and a uniform distribution with equal variance.

Definition 4.2. (Signal-to-Noise Ratio [58, Equation 2])
The signal-to-noise ratio is the proportion σ_M^2/σ_E^2, where σ_M^2 is the variance of the signal (e.g. the manifold) and σ_E^2 is the the variance of the noise.

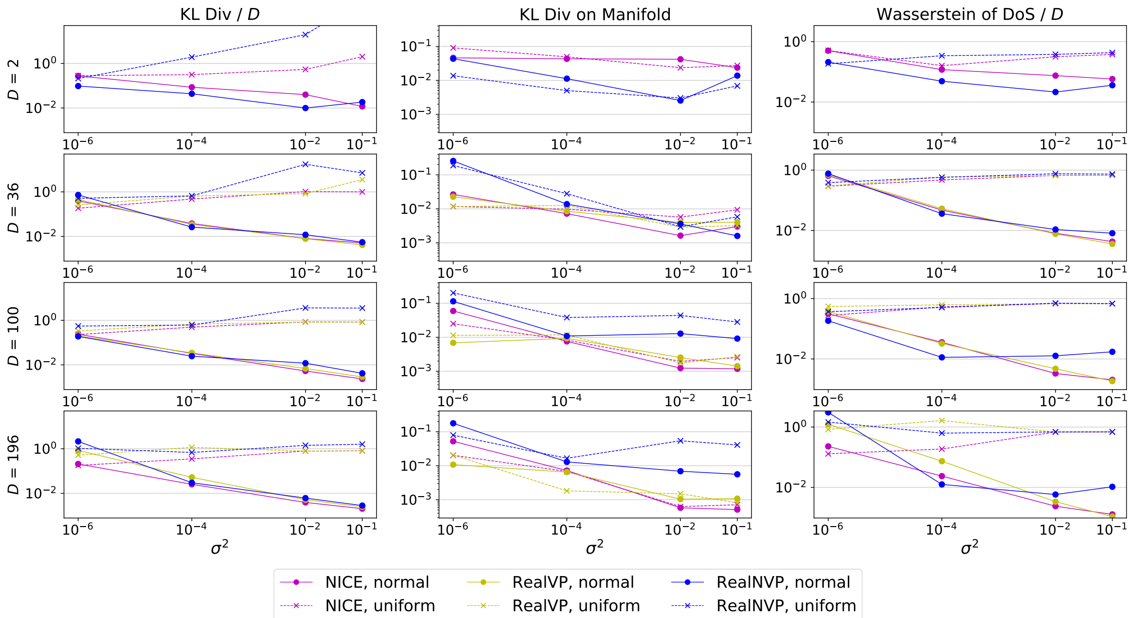

Figure 4.10: **Uniform and normal noise comparison, qualitative.** Models q trained on the Smooth Step distributions with uniform noise (dashed lines) and normal noise (solid lines) used for inflation. "KL Div / D" is the KL divergence from q to the inflated distribution p. "KL Div on Manifold" is the KL divergence from q along the manifold, to the ground truth manifold distribution. "Wasserstein of DoS" is the Wasserstein distance of the DoS's of q and p. The columns show the embedding distribution D and the x-axes show the noise levels used for inflation. For $D = 2$, RealVP coincides with NICE. For all manifolds, see Appendix A.7.

Our hypothesis makes statements about the ability of VP models to approximate an inflated manifold depending on the type of inflation noise, but not for NVP models or the approximation of the manifold. However, since we can measure the performance of the NVP model and for the approximation of the manifold, we will show those results and briefly discuss them in subparagraphs after we discussed the performance of the VP models.

VP Models

The first column in Figure 4.10 shows that the VP models (i.e. NICE and RealVP) perform better on the inflated manifold when normal noise is used. As predicted from our hypothesis, this depends on the DoS which was also approximated better when normal noise is used (third column, Figure 4.10). This confirms Hypothesis 3.3, which states that degenerate-DoS noise should not be used for inflation for VP normalizing flows. From the KL divergence to the inflated manifold and the Wasserstein distance of the DoS's in Figure 4.10 (first and thrid column), it can be seen that for small noise levels (i.e. x-axes at $\sigma^2=10^{-6}$), the performance with normal and uniform noise is often similar, whereas for higher noise levels normal noise is always better. However, in Section 4.2.1 and 4.2.2 we find that the models used generally perform poor at smaller noise levels. Therefore, the similarity for normal and uniform noise at small noise levels may indicate a general poor behavior of our models with small noise levels and not an inaccuracy of our hypothesis.

RealNVP

Even though RealNVP can change its DoS and therefore should perform comparable with normal and uniform inflation noise, it also performs better with normal noise (first and third column, Figure 4.10). Why this happens is not clear, though a reason could be that the model used is not expressive, i.e. deep, enough to transform uniform inflation noise into the normal prior, even though we use a deeper version than in the RealNVP paper[1]. Since RealNVP performs better when it has to map normal inflation noise into the normal prior, our results also imply that the use of prior matching noise is favorable for NVP normalizing flows.

Along the Manifold

Our hypothesis makes no statement about the performance along the manifold. From middle column "KL Div on Manifold" in Figure 4.10, it can be seen that the performances are rather similar, although normal noise is still better. While this also argues for the use of prior matching noise, it shows that there seems to be an innate behavior in normalizing flows that prioritizes modeling the signal (i.e. the manifold) over the noise, otherwise we would observe a divergence similar to that of the inflated manifold (cf. left column "KL Div / D" to middle column "KL Div on Manifold" in Figure 4.10).

[1] We use 10 coupling layers to train on a simple distribution, while Dinh et al. [2, Section 3.6 and 4.1] use 9 coupling layers for CIFAR-10, which is a more complex dataset than ours.

4.2.5 Summary and Discussion

In this subsection, we summarize the findings of this section and give further discussions.

We confirmed Hypothesis 3.1 and 3.2, which states that a volume change is necessary for a good approximation of an inflated manifold in a large $\frac{d}{D}$ regime, and that it is not necessary for small $\frac{d}{D}$, respectively, in Section 4.2.1. The results were consistent with different VP transformations, i.e. additive and VP affine. This also shows that our model based on VP-affine transformations works and should be considered when choosing between different VP transformations. Our experiments also showed that VP normalizing flows often outperform the NVP normalizing flows in the small $\frac{d}{D}$ regime for the approximation of the inflated manifold (Section 4.2.1) and along the manifold (Section 4.2.2). Since VP normalizing flows are less expressive than NVP normalizing flow (Section 3.3), this could be explained by Occam's razor, which has been shown to lead to better Bayesian models [19]. The poor performance of NVP normalizing flows along the manifold manifests itself in nonessential mass concentrations (see Figure 4.6). Since this behavior is unexpected, it will be investigated in the next section. Until the problematic behavior of NVP normalizing flows is solved, we suggest to use NVP normalizing flows for large $\frac{d}{D}$ and VP normalizing flows for small $\frac{d}{D}$. Note that in order to increase D and thus decrease $\frac{d}{D}$, one could pad the data with empty dimensions, although research is needed to verify the benefits of this technique. We note that even though we tested our hypothesis for multiple manifolds and inflation noise levels, we only tested one ratio for the large $\frac{d}{D}$ case (i.e. $\frac{1}{2}$)[1].

We also confirmed Hypothesis 3.3 which states that degenerate-DoS noise should not be used for VP normalizing flows in Section 4.2.4. we found that this is also beneficial for NVP normalizing flows. However, NVP normalizing flows can change their DoS and thus adapt to different types of noise, while VPs cannot (see Section 3.3). We only compared noise that is commonly used, i.e. normal and uniform, and one type of prior, i.e. a normal distribution. However, our theoretical analysis also suggests that the more similar the DoS of the inflation distribution is to the prior distribution, the better VP normalizing flows should perform, which is an interesting direction for future research (see end of Section 3.4.1).

Since the true distribution of common datasets is usually unknown, the log-likelihood for the inflated manifold is often chosen as a performance indicator [2, 10, 11, 21, 33, 59, 60]. Although the maximum log-likelihood minimizes the KL divergence between the target and model distributions [31], and thus the (deflated) manifold distribution should be matched as well, our results from Section 4.2.3 show that a better log-likelihood of a model, is not a sufficient condition for a better approximation of the manifold distribution. Therefore, if the goal is to obtain good likelihoods for the manifold, on can generally rely on the likelihood of RealNVP for the inflated manifold to predict a good manifold approximation when $\frac{d}{D}$ is large, and on VP models if $\frac{d}{D}$ is small.

We found one comparison between VP and NVP normalizing flows which we can relate our work to. Kingma & Dhariwal test how well an NVP and a VP implementation of their model approximates one distribution [11, Figure 3]. The NVP implementation performed better than the VP implementation. This could be because they used uniform noise for inflation, because performances can vary for different datasets and noise levels, which is evident throughout our experiments (see Appendix A.3), or because they compared additive to affine transformation in-

[1]We did this to better observe the learned pdf along the manifold, which is impractical on paper when the dimensionality of the manifold is greater than 1.

stead of affine to VP-affine transformations, which are more expressive VP normalizing flows, as shown in Appendix A.2. However, it could also be because they use natural data that have a more complicated embedding and distribution than our data.

Even though our goal is not to find good noise levels, our data suggests that all models perform better the stronger the (prior distribution matching) inflation noise (best seen in Appendix A.3). This is consistent with the results of Horvat and Pfister [23, Figure 4, bottom right]. It is an interesting direction for future research to find noise levels which improve the approximation but do not smooth the manifold beyond recognition.

4.3 Exploring RealNVPs Behavior

Since additive and VP-affine transformation are a proper subset of affine transformations[1], RealNVP is at least as expressive as the VP models we use (i.e. NICE and RealVP). However, the results in Section 4.2.2 show that the VP models often perform better than RealNVP in the regime of small $\frac{d}{D}$ along the manifold, in terms of KL divergence. The mismatch between the learned distribution of RealNVP with the ground truth distribution is most evident in unexpected mass concentrations in from the learned pdf along the manifold (Figure 4.6). In this section, we will explore the reason for this behavior.

Our line of argument is: first, we show that the volume element is responsible for the poor performance along the manifold. While we ensure that this is not caused by a poor local minimum, we note that the pdf along the manifold fluctuates significantly during training. We than find that RealNVP strongly changes its weights during training and finally attribute this to the steep loss landscape of RealNVP.

Performance with and without volume element

We follow the path of Nalisnick et al. [9] and investigate the performance of RealNVP with and without the volume element. A model that uses the transformation of RealNVP and omits the volume element requires a normalizing constant. Therefore, we refer to this model as *Real Normalizing Constant* (RealNC). In Figure 4.11 we show the KL divergence from RealNVP and RealNC to the distribution of the inflated manifold, and in Figure 4.12 we show the KL divergence from RealNVP and RealNC along the manifold to the manifold distribution.

Definition 4.3. (Real Normalizing Constant) A model which uses RealNVPs transformation f_{RealNVP} but not its volume element is RealNC with pdf

$$\begin{aligned} q_{\text{RealNC}}(x) &= \frac{1}{Z}\, p_Z(\, f_{\text{RealNVP}}(x)\,), \\ \text{where } Z &= \int_{\mathcal{X}} p_Z(\, f_{\text{RealNVP}}(x)\,)\, dx. \end{aligned} \tag{4.5}$$

[1]For $s(\cdot) = 0$, cf. transformation 2.7 of RealNVP to transformation 2.10 and 2.12 of the VP models.

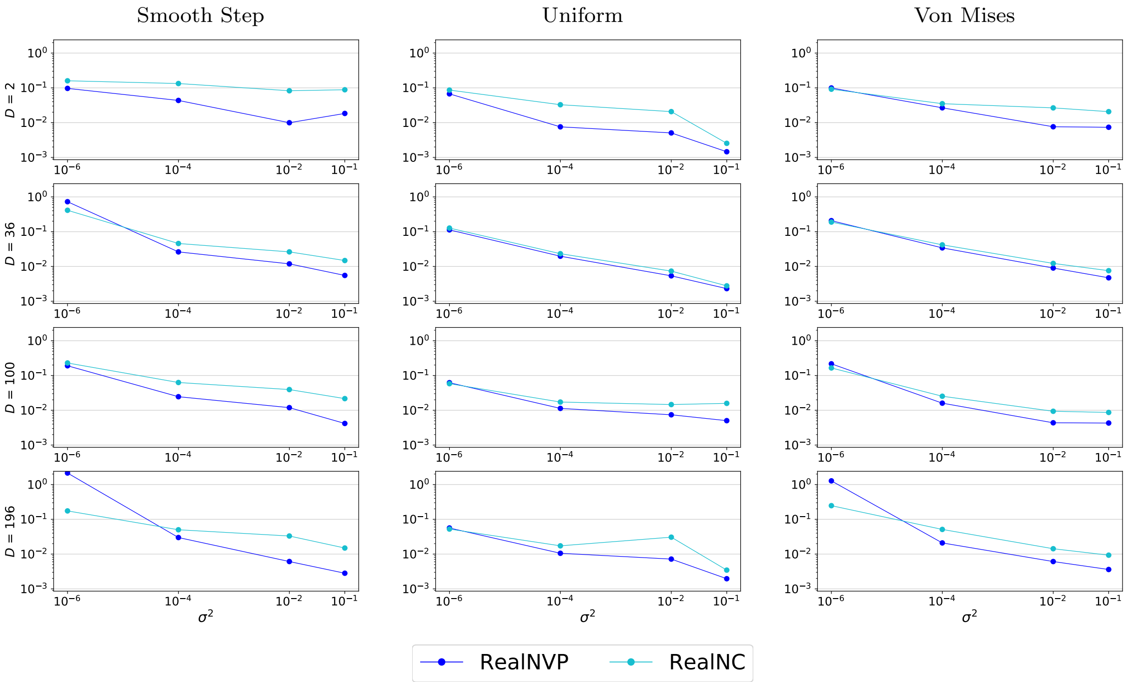

Figure 4.11: **KL divergence for RealNVP and RealNC to the inflated distribution**. RealNVP q_{RealNVP} is trained on samples of the inflated manifold p_X. RealNC is the distribution of q_{RealNVP} without its volume element. The dots show the KL divergence from RealNVP and RealNC to p_X. Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale.

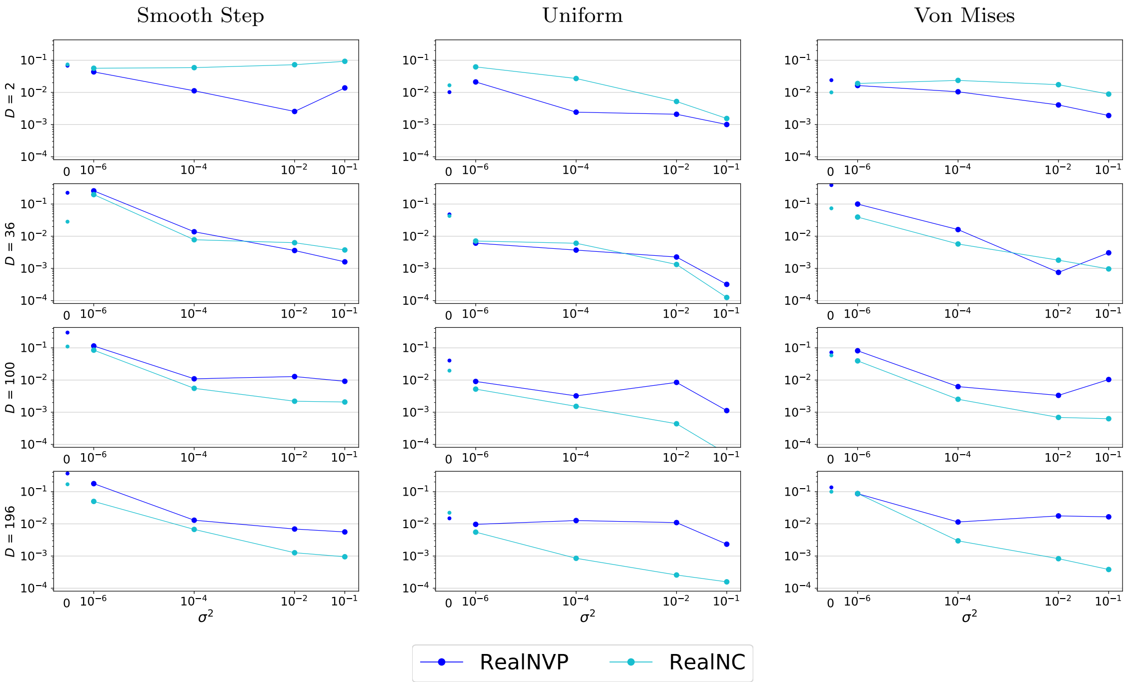

Figure 4.12: **KL divergence for RealNVP and RealNC along the manifold**. RealNVP q_{RealNVP} is trained on samples of the inflated manifold p_X. RealNC is the distribution of q_{RealNVP} without its volume element. The dots show the KL divergences from the distribution of RealNVP and RealNC along the manifold to the manifold distribution. Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale.

From Figure 4.11, we observe that RealNVP approximates the inflated manifold better most of the time. This is not surprising, since RealNVP is trained to perform well on the inflated manifold, while RealNC is a mere byproduct of RealNVP. However, for small $\frac{d}{D}$, RealNC generally outperforms RealNVP along the manifold as can be seen in Figure 4.12 (bottom rows, $D = 36, 100, 196$). In this regime RealNVP also performs notably worse than the VP models along the manifold (cf. Figure 4.5). Considering the change of variables formula that governs RealNVP, i.e. $q_{\text{RealNVP}}(x) = p_Z(f(x))\, v_f(x)$, the transformation appears to work fine, since RealNC represents the left term, i.e. $p_Z(f(x))$, while the volume element $v_f(x)$ is causing problems.

We confirm this by comparing the actual volume element v_f and the optimal volume element v^*, that would lead to the true density p^* under the learned transformation f of RealNVP ($p^*(x) = p_Z(f(x))\, v^*(x|f)$), in Figure 4.13. We observe that, while v_f works acceptably for $D = 2$ (first row, Figure 4.13), it mostly diverges by orders of magnitude[1] from the optimal volume element for $D = 36, 100, 196$ (bottom rows, Figure 4.13).

Additionally, we note that the volume element can be made responsible for the unexpected hills that appear at some sides of the learned pdf along the manifold, e.g. cf. $m \in (-6, -4)$ for cell $D = 100$ and $\mathcal{N}(0, 10^{-1})$ in the volume element in Figure 4.13 to the learned pdf of RealNVP along the manifold in Figure 4.6.

[1]Note that the volume elements in Figure 4.13 are log-scaled.

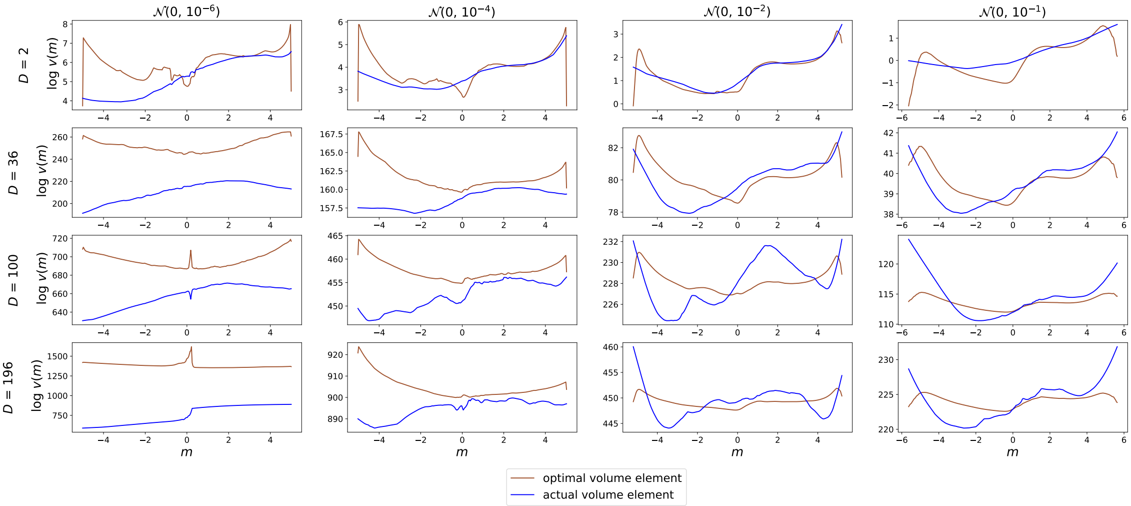

Figure 4.13: **Actual and optimal volume element comparison** for the Smooth Step distributions along the manifold, $m \in \mathcal{M}$. The learned logged volume change $\log v_f$ along the manifold is labeled "actual volume change". The logged volume change $\log v^*$ that would lead to the true density p^* given the learned transformation f is labeled "optimal volume change" ($p^*(x) = p_Z(f(x))\, v^*(x|f)$). Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation.

Pdf during training

To see if we coincidentally stop at a bad local minimum, we run the training without early stopping for 1000 epochs for an exemplary distribution. We show the pdfs at the end of the training in Figure 4.14, and observe that the VP models tend to converge to their final pdfs, while RealNVP still changes them significantly. This change in pdf is visible throughout the training and for other target distributions.

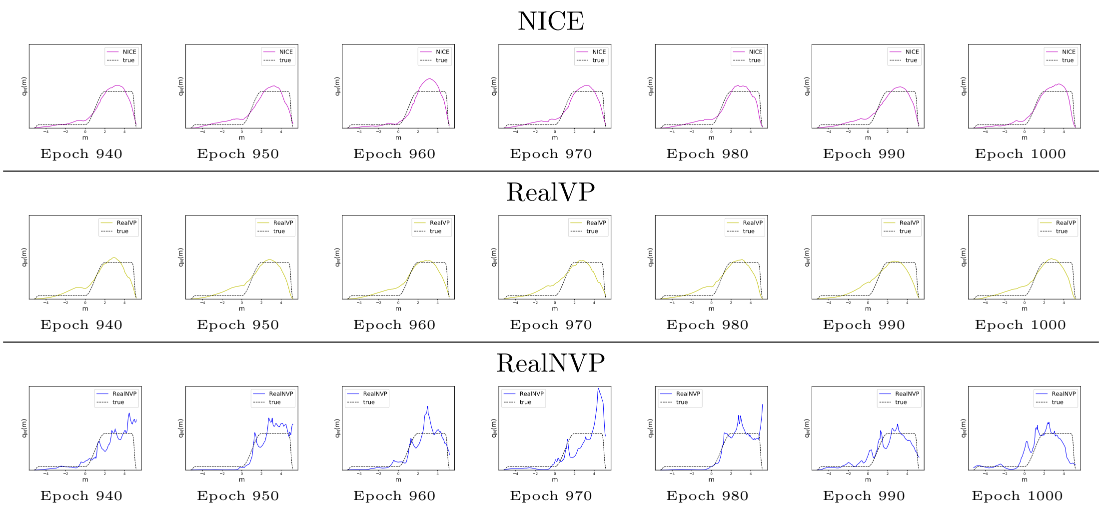

Figure 4.14: **Changes of the learned pdf q_M along the true manifold** ($m \in \mathcal{M}$) at the end of the training for the Smooth Step distribution in 100 dimensions and inflation noise $\mathcal{N}(0, 10^{-2})$. Pdfs are plotted every 10 epochs.

Weight changes

This change in pdf would indicate a strong loss fluctuation. However, the loss varies in a similar way for all models (right column in Figure 4.15). Therefore, we assume that RealNVP has a stronger weight change than the VP models. We test this by computing the *relative weight change*, which we show in Figure 4.15.

Definition 4.4. (Relative Weight Change [61, Equation 1]) The relative weight change is

$$\frac{\|w^t - w^{t-1}\|_1}{\|w_i^{t-1}\|_1},$$

where w^t is the weight vector of a normalizing flow at epoch t and $\| \cdot \|_1$ is the L_1 norm.

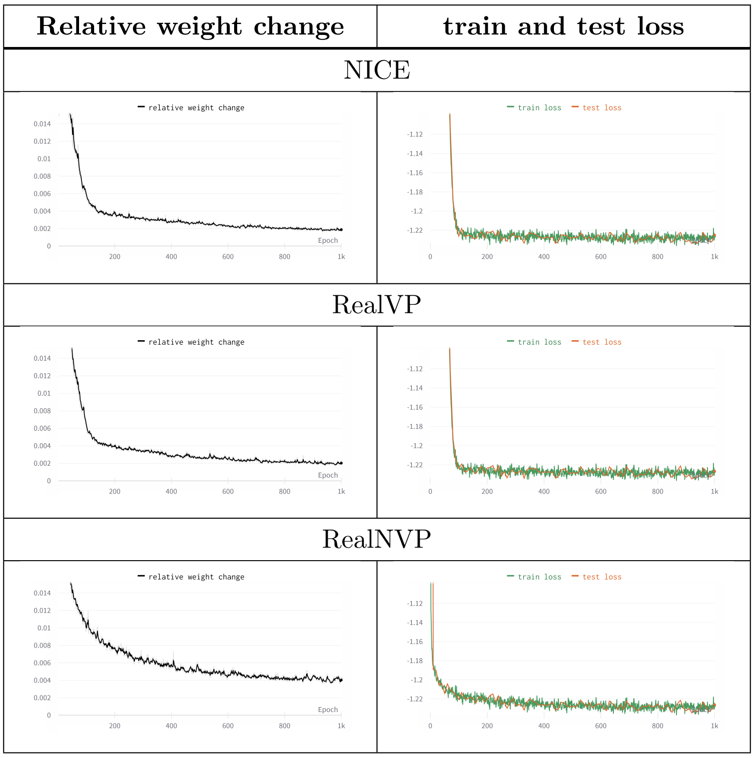

Figure 4.15: **Weight change and loss during training** for the Smooth Step distribution in 100 dimensions with inflation noise $\mathcal{N}(0, 10^{-2})$. X-axes show the epoch. Left: Smoothed relative weight change. Right: Training loss in green, test loss in orange.

We observe that while the losses of all models converge (right side in Figure 4.15), the change in weight of RealNVP is more than twice as high as that of the VP models (left side at epoch 1K in Figure 4.15). It is notable, that RealNVP reaches a loss that is about 2%[1] away from its final loss after about 20 epochs (bottom right cell, Figure 4.15, loss at epoch 20), while NICE and RealVP reach their convergence level after about 100 epochs (top right cells, Figure 4.15, loss at epoch 100). Since RealNVP convergence faster and its weights fluctuate more, we assume that its loss landscape significantly differs from that of the VP models, and therefore plot it.

[1]Talking about loss differences in percent might not be meaningful, but we have reference it in some way.

Loss landscape

Figure 4.16 shows the loss landscape around the converged parameters. For this visualization we use the method and implementation from [62], which evaluates the loss of the training set for a model, when we change its weights along a randomly rotated 2D plane. The center of this plane is on the converged weights. We move along this plane such that the motion on each layer is normalized by its length.

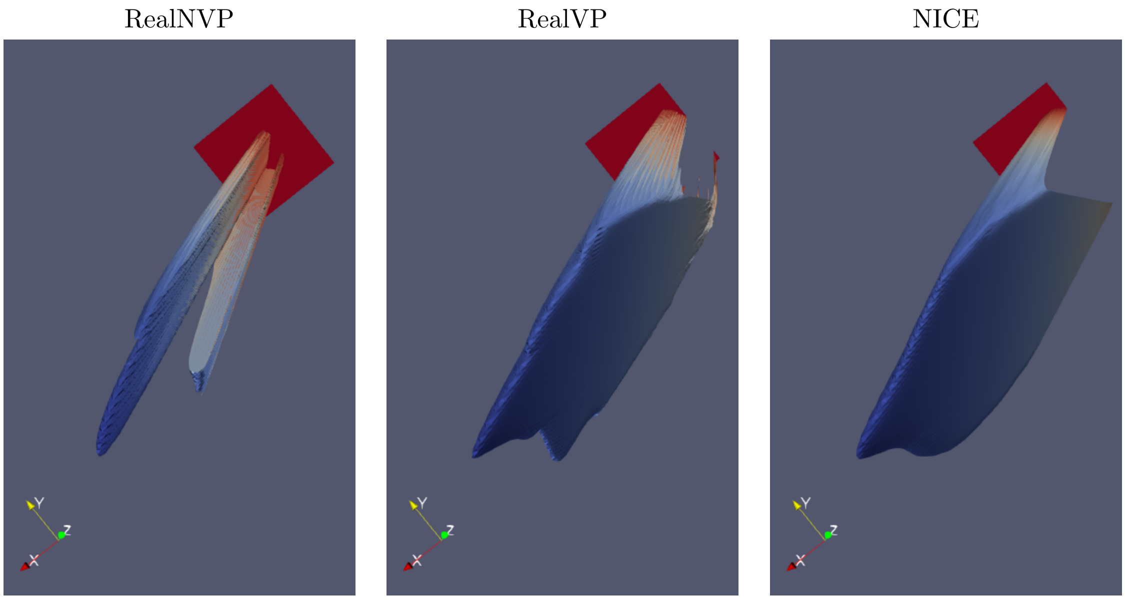

Figure 4.16: Loss surfaces centered at the point of convergence seen from below for the Smooth Step distribution in 100 dimensions with inflation noise $\mathcal{N}(0, 10^{-2})$. Blue is a better loss, red worse. The loss is upper bounded at $\hat{l} - 10 \cdot |\hat{l}|$ where $\hat{l}$ is the best loss in the visible area.

It is noticeable that the loss surface of RealNVP is much steeper in every direction than that of the other models. Since a steep loss space implies a steep gradient, this can explain why RealNVP reaches its loss convergence more rapidly than the VP models, which have to move down along the slope of the right side more slow. The VP models also have a steep region on the left side, however we can assume that there are more such slopes in other directions of this high dimensional space[1].

The steep slope of RealNVP can also explain the weight fluctuation, when considering that a small change in weight can lead to a location in the parameter space with a strong gradient, which than leads to strong change in weight. The VP models also have areas of high gradients, but a significant amount of the loss space has moderate gradients which does not cause a strong weight change.

Given the steep loss landscape of RealNVP and the fact that the VP models often outperform RealNVP, we assume that the loss space of the VP models is better conditioned, leading to a more desired minimum, i.e. one that better matches the (deflated) manifold distribution (as shown in the previous Section 4.2). Since smoother loss spaces have been shown to lead to better convolutional neural networks for classification [62], investigating how to better condition the loss space of

[1]The models used have around 4,400,000 parameters for $D = 100$.

RealNVP seems like a promising avenue. However, observing and understanding high-dimensional spaces is complex and thus interpretations should be taken with caution.

4.3.1 Learned Transformations and Probability Density Functions

Although RealNVP can change its DoS and therefore should be able to approximate the target distribution well, we still observe some flaws even in the regime of large $\frac{d}{D}$ (i.e. the regime where RealNVP performs better than the VP normalizing flows as shown in Section 4.2). To gain a deeper understanding of the mechanisms of the learned transformation, we present the pdf of the two-dimensional space and additional information in Figure 4.17 and a detailed visualization of the transformation in Figure 4.18. They are explained separately after the figures. We point out that while these figures reveal interesting features, they often do not explain why something happens, but suggest interesting avenues for future research.

To describe the visual phenomena we observe, we will use an informal language in this section. Moreover, since the goal of a model is to transform the inflated manifold so that its distribution is represented by latent distribution (and the volume element), we will speak of the transformation as if it transformed the inflated manifold instead of the data space[1]. We do not show RealVP because it coincides with NICE for $D = 2$.

[1]For example, we say "a model transformed the manifold to the center of the latent distribution" instead of "a model transformed the space in which the ground truth of the manifold resides to the center of latent distribution".

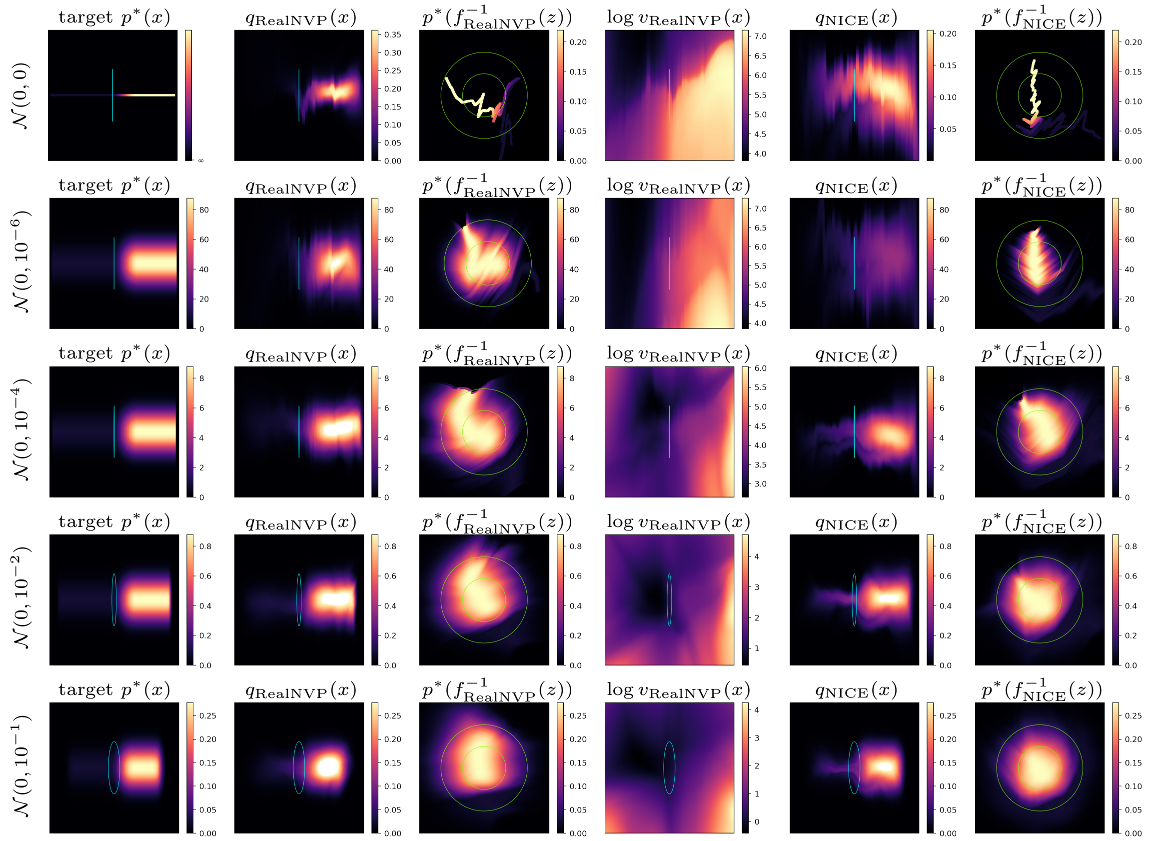

Figure 4.17: Pdf, transformation and volume element plots of the Smooth Step distributions in two-dimensional space. The rows show the different noise levels $\mathcal{N}(0, \sigma^2)$ used to inflate the manifold for training. The plots in the data spaces (first, second, fourth and fifth column) are stretched on the y-axis. The stretch is represented as stripes or ellipses, which in the none-stretched space are circles with radius 2σ if $\sigma \neq 0$, otherwise $2 \cdot 10^{-3}$. The embedding of the manifold (i.e. the rotation) is not shown. **First column** target $p^*(x)$ shows the ground truth distribution of the inflated manifold. **Second and fifth columns** $q(x)$ show the learned pdfs with the color scale of $p^*(\cdot)$. When this scale is exceeded, white is drawn (e.g. $q_{\text{RealNVP}}(x)$ middle row, $\mathcal{N}(0, 10^{-4})$). **Third and sixth column** $p^*(f^{-1}(z))$ are in the latent space and show the learned mapping of the ground truth distribution to the latent space (the color scheme of $p^*(x)$ remains). The green circles have radius 1 and 2 and contain **Fourth column** $\log v_{\text{RealNVP}}(x)$ shows the logged volume element of RealNVP in the data space.

In the following paragraph we refer to Figure 4.17. For the case of low noise, it can be seen that NICE barely matches the target distribution, while RealNVP already performs acceptably (cf. second column "$q_{\text{RealNVP}}(x)$" and fifth column "$q_{\text{NICE}}(x)$ to the first column "target $p^*(x)$", first three rows, $\mathcal{N}(0, \{0, 10^{-6}, 10^{-4}\})$). As the noise increases, this effect disappears (bottom two rows, $\mathcal{N}(0, \{10^{-2}, 10^{-1}\})$) and RealNVP and NICE perform comparably, as reflected by the KL divergences (Figure 4.2 and 4.5). We do not know why this changes with different noise levels, but finding good noise levels is not an objective of this work.

From the transformations to the latent space (column three and six, "$p^*(f^{-1}(z))$") we find that RealNVP bends the manifold to "use as much probability" in the latent space as possible (well visible in middle row, $\mathcal{N}(0, 10^{-4})$). It seems that NICE puts the area of highest probability of the manifold in the area of highest probability of the prior, i.e. the center (well visible in second row, $\mathcal{N}(0, 10^{-6})$). In the next figure we will see and discuss more details of this transformation. Overall, this depiction gives the intuition that a model needs to transform the data distribution, in such a way that it "collects" as much probability in the latent space as possible, in other words the green circles should be filled with the manifold. It is evident that both models do so (note that there is almost always small probability in the outer green ring seen as light blue color).

Since noise is added symmetrically, one expects a rather symmetric learned distribution. However, the learned distributions for both models (column two and five, "$q(x)$"), show asymmetrical patterns. This can be explained by the transformations in column three and six ("$p^*(f^{-1}(z))$"), where RealNVP bends the distribution so that one side of the data obtains more probability than the other, e.g. upper part on the left and the right side between the bright part in middle row ($\mathcal{N}(0, 10^{-4})$), third column. NICE, on the other hand, often transforms the inflated manifold not quite to the center, resulting in asymmetry, well seen in the second row $\mathcal{N}(0, 10^{-6})$, column six. We assume that this happens, such that the tail of the target distribution with low probability (left side of the target $p^*(x)$, first column), can still "collect" some probability (sixth column, "$p^*(f^{-1}_{\mathrm{NICE}}(z))$", first two rows, $\mathcal{N}(0, \{0, 10^{-6}\})$).

The volume element of RealNVP (column five, "$\log v_{\mathrm{RealNVP}}(x)$") shows large differences in strength, e.g. $v_{\mathrm{RealNVP}}(x)$ ranges from about $e^4 \approx 45$ to $e^7 \approx 1100$ for $\mathcal{N}(0, 10^{-6})$, second row. It can be observed that the volume element increases, the farther it is from areas of significant probability (in our case, the more it goes to the edges of the plot). We assume that this happens so that areas farther from the manifold (i.e. of low probability) still receive some probability, e.g. the right side in the plot of the fourth row ($\mathcal{N}(0, 10^{-2})$), second column ("$q_{\mathrm{RealNVP}}(x)$") has very low probability, but the volume element for this region still leads to some probability.

At last, we mention the special case without inflation noise (first row, $\mathcal{N}(0, 0)$). We learn from a set of finite samples, which is thus a discrete distribution. A model could learn this distribution by placing high density peaks around each sample [21]. However, from column three and six ("$p^*(f^{-1(x)})$") it is clear that this does not happen and instead a smooth transformation is learned. The transformation itself is curious, since it does not use all areas of high density in the latent space. We will discuss this in detail using the next figure.

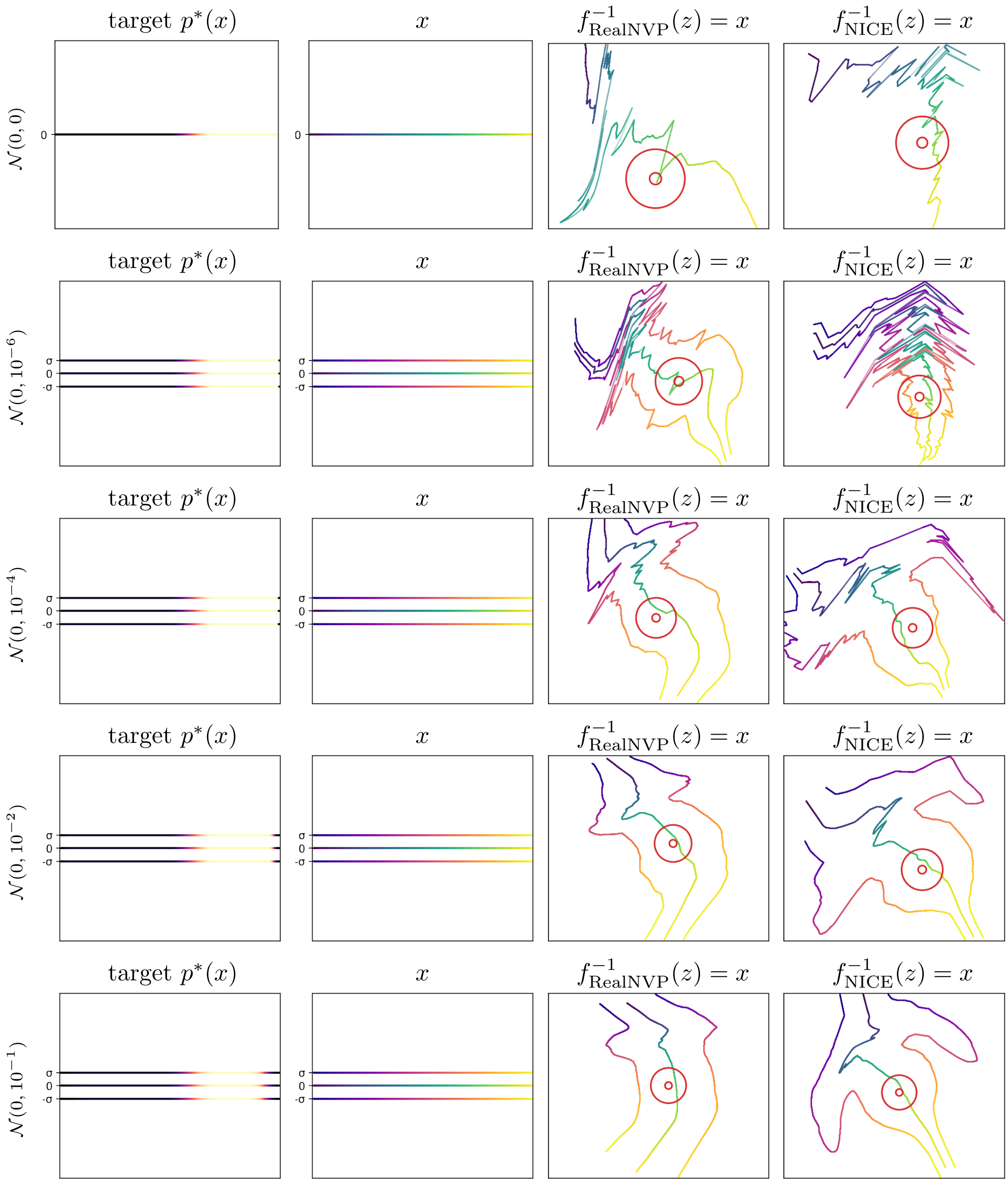

Figure 4.18: Color-coded transformation of the Smooth Step distributions in two-dimensional space. The rows show the noise levels $\mathcal{N}(0, \sigma^2)$ used to inflate the manifold for training. **First two columns** show the manifold and two lines that are one standard deviation σ away from the manifold in both direction. The embedding of the manifold (i.e. the rotation) is not shown. **First column** target $p^*(x)$ is the ground truth density for reference. Brighter means higher density. **Second column** is the color coding used to show the transformation to the latent space. **Third and fourth column** are the learned transformations in the latent space from RealNVP and NICE applied to the color-coded x. The read circles are centered and have radius 0.1 and 0.5.

In the following paragraph, we refer to Figure 4.18. The transformation strategy for both models with small inflation noise to "fill out the latent space" seems to be a zigzag pattern (third and fourth column ("$f^{-1}(z) = x$"), first three rows, $\mathcal{N}(0, \{0, 10^{-6}, 10^{-4}\})$). This zigzag explains the ups and downs of the learned pdf along the manifold which we observe for all models and all datasets (Figure A.5, columns $\mathcal{N}(0, \{0, 10^{-6}, 10^{-4}\}))$ since a movement towards the center increases the density (i.e. the "zig") and a movement away decreases it (i.e. the "zag"). In the absence of noise, it is worth noting that while both models could produce an extreme zigzag pattern (as seen in the second row, $\mathcal{N}(0, 10^{-6})$), which should result in better divergence, they do not.

For high noise levels (bottom two rows, $\mathcal{N}(0, \{10^{-2}, 10^{-1}\})$), this zigzag pattern disappears, leading to smoother pdfs also reflected in the learned pdf plots in Figure A.5 (two rightmost columns, $\mathcal{N}(0, \{10^{-2}, 10^{-1}\})$).

Since the probability on an annulus in the latent space decreases as distance to the center increases, we expected a spiral-like transformation. Instead, we observe either a zigzag patterns that could be caused by ReLu activation, leading to sharp decision boundaries (c.f. [63], where it is shown that neural networks are decision trees), or almost straight transformations. It may be of interest to support normalizing flows in the creation of spiral-like transformations.

Chapter 5

Conclusion

Modeling complex high-dimensional distributions is an important task. A good candidate for solving this problem is the model class of normalizing flows. They can be categorized into volume-preserving (VP) and non-volume-preserving (NVP). For our problem setting, the distribution to be learned emerges from a d-dimensional manifold and noise, which is added to inflate the manifold such that the dimensionality D of the normalizing flow is matched. In this thesis, we answered the question if the ability to change the volume is necessary to model a distribution well. We linked this question to the dimensionality of the manifold d, the dimensionality of the data D, and the type of noise used to inflate the manifold.

We also took a closer look at the approximation of the manifold distribution and investigated an unexpected behavior of an NVP normalization flow.

For theoretical analysis, we introduced the density of states (DoS) to the domain of deep generative models. This tool can be used to analyze whether a model has the necessary (density) states to match the states of a target distribution. We used it to analyze the difference between VP and NVP normalizing flows and derived that a change in volume (i.e. a change in states) is not necessary when $\frac{d}{D}$ is small and the inflation noise matches the distribution of the prior of the normalizing flow, and the a volume change is necessary when $\frac{d}{D}$ is large or the type of the inflation does not match the prior of the normalizing flow.

We validated our theoretical results experimentally. To exclude the possibility that NVP normalizing flows perform better than VP normalizing flows because there transformation is more expressive by design, i.e. standard NVPs use affine transformations [2, 11], while standard VPs use additive transformations [1], we introduced VP affine transformations (Section 2.2.3), which we used along with additive transformation based normalizing flows on the VP side. As datasets we used manifolds with known ground truth distribution, such that we are able to measure how well the density of the manifold is matched. We found that VP normalizing flows perform comparable to NVP normalizing flows when $\frac{d}{D}$ is small for the approximation of the inflated manifold, and that they perform better for the approximation of the (deflated) manifold distribution. Since we also found that a better approximation of the inflated distribution does not imply a better approximation of the manifold distribution, we suggest to use VP normalizing flows for small $\frac{d}{D}$. However, we used artificial data, to have manifolds with known ground truth distribution. Since this cannot reflect the complexity of natural data, and since the approximation of the distribution of the manifold should be a key indicator for choosing one model over another, we suggest to develop natural data sets with known ground truth distribution, to have a proper benchmark for the approximation of the density of the manifold.

Lastly, we investigated why NVP normalizing flows perform worse than VP normalizing flows even though they are more expressive. We found that NVP normalizing flows have a much steeper loss landscape, which leads to unstable training. Better conditioning of the loss space of NVP normalizing flows is a promising direction for future research. We also found that the transformations from the data to the latent space of normalizing flows follow a zigzag pattern. To take advantage of the spherically decreasing pdf of the latent distribution, we should investigate how to induce spiral-like transformations.

Although normalizing flows have the potential to model complex high-dimensional distributions, a one-model-fits-all solution seems unlikely and further research is needed to solve problems with normalizing flows. In conclusion, we answer the question from the title of this thesis with: *It depends - but more often than not, the volume can be preserved.*

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Appendix A

Appendix

A.1 Density of States Related

A.1.1 Location Independence of the Density of States

Let $A \sim F$ and $B = A + c$ for some c of the space of A, then

$$\log p(B) = \log p(A + c) = \log p(A).$$

Thus, the DoS of A is independent of its location.

A.1.2 Density of States of Gaussian Distributions

Let $B \sim \mathcal{N}(\mu, 1)$ and $A := \sigma B$, such that $A \sim \mathcal{N}(\mu, \sigma^2)$.
The pdf of A is $p(a) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(a-\mu)^2}{\sigma^2}}$.
We are interested in the density of $\log p(A)$. Since the DoS is location-independent (see Appendix A.1.1), we set $\mu = 0$.

$$
\begin{aligned}
&P(\log p(A) \leq c) && \text{(A.1)}\\
=&P(\log \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{A^2}{\sigma^2}}\right) \leq c) && \text{(A.2)}\\
=&P(-\underbrace{\log(\sqrt{2\pi\sigma^2})}_{\phi} - \frac{1}{2}\frac{A^2}{\sigma^2} \leq c) && \text{(A.3)}\\
=&P(-\frac{1}{2}\frac{A^2}{\sigma^2} \leq c + \phi) && \text{(A.4)}\\
=&P(-\frac{(\sigma X)^2}{\sigma^2} \leq 2(c + \phi)) && \text{(A.5)}\\
=&P(-X^2 \leq 2(c + \phi)) && \text{(A.6)}\\
=&\int_{-\infty}^{2(c+\phi)} p_{\chi^2}(-t)\, dt, \quad p_{\chi^2} \text{ is the pdf of the } \chi^2 \text{ distribution} && \text{(A.7)}\\
=&\int_{-\infty}^{c} p_{\chi^2}(-2(s + \phi)) \cdot 2\, ds, \quad \text{by change of variables} && \text{(A.8)}\\
=&\int_{-\infty}^{c} 2 \cdot p_{\chi^2}(-2s - 2\log(\sqrt{2\pi\sigma^2}))\, ds && \text{(A.9)}
\end{aligned}
$$

$$= \int_{-\infty}^{c} 2 \cdot p_{\chi^2}(-2s - \log(2\pi\sigma^2)) \, ds \, . \tag{A.10}$$

Thus the pdf at point c of $\log p(A)$ is

$$p_{\log p(A)}(c) = 2 \cdot p_{\chi^2}(-2c - \log(2\pi\sigma^2)). \tag{A.11}$$

To show correctness, we plot the sampled and derived DoS in Figure A.1.

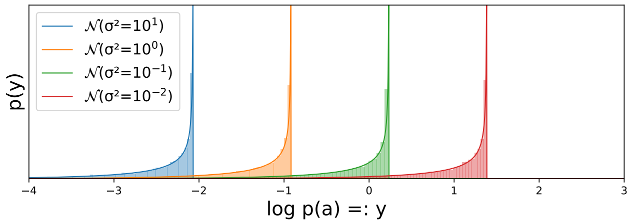


Figure A.1: Histogram is the sampled DoS, solid line is the derived DoS of different Gaussians.

A.1.3 Translation Equivariance of Convolution

Let f and g be two functions. Let δ_t be the Dirac delta function at position t, then a convolution of f and δ_t translates f by t [53].
Since convolution is associative and commutative, it holds that

$$(\delta_t * f) * g = f * (\delta_t * g) = \delta_t * (f * g). \tag{A.12}$$

□

A.1.4 Form of the Density of States of Isotropic Gaussian Distributions

Given a dimensionality D, let $A = (A_1, ..., A_D) \sim \mathcal{N}(m, \sigma^2 I_D)$ be an isotropic D-dimensional Gaussian with DoS $p_{\log p(A)}$.
Since $p_A(a) = \prod_{i=1}^{D} p_{\mathcal{N}(m_i,\sigma^2)}(a_i)$ due to the independence of all A_i from each other, the DoS can be expressed as

$$\log p(A) = \sum_{i=1}^{D} \underbrace{\log p_{\mathcal{N}(m_i,\sigma^2)}(A_i)}_{B_i} = \sum_{i=1}^{D} B_i. \tag{A.13}$$

The pdf of the sum of the random variables $\{B_i\}$ can be computed by convolution (Equation 3.3), and a single summand B_i translates with different variance σ^2 and preservers its shape (Lemma 3.2). Since convolution is translation equivariant (Lemma 3.2), $\log p(A)$ translates with varying σ^2 and preservers its shape.

A.1.5 Density of States of High-Dimensional Uniform Distributions

Let $A \sim U^D[-a, a]$ be a D-dimensional uniform distribution. The DoS of A is $\log p(A)$ and

$$\log p(A) = \log(\frac{1}{2a})^D = -D \log 2a. \tag{A.14}$$

Thus the entire mass of $\log p(A)$ resides at $-D\log(2a)$ and thus its pdf is expressed as $\delta(x + D\log 2a)$.
□

A.1.6 Inflation Noise Levels only Translate the DoS

Consider Equation 3.9, which describes the DoS of a Q-normally reachable manifold inflated with noise, i.e.

$$\log q_N(X) = \log p(E) + \log p_M(M), \tag{A.15}$$

for $X = E + M$, where $M \sim p_M$ and E is distributed according to some noise. Also $\log p(E)$ and $\log p_M(M)$ are random variables distributed according to the DoS of E respectively M. For some noise distributions, different variances translate the DoS of E by τ_t while preserving its shape (e.g. Gaussian or uniform, see Figure 3.2 respectively Appendix A.1.5). Since Equation A.15 is the sum of independent random variables, its pdf can be computed by convolution (see Equation 3.3). Since convolution is translation equivariant (Property 3.1), the pdf of $\log q_N(X)$ translates with τ_t for different noise levels while preserving its shape.
□

A.1.7 DoS of Inflated Manifolds become Limiting Gaussians

In this section, we show how the shape of the DoS becomes a limiting Gaussian distribution, when $\frac{d}{D} \to 0$. To show this, we make use of the "Central Limit Theorem for Dependent Random Variables" [64], which we briefly summarize.

The Central Limit Theorem for Dependent Random Variables
Given a sequence of random variables

$$Y^n = \{X_1, X_2, ..., X_n\}, \tag{A.16}$$

with $\mathbb{E}[X_i] = 0$ and $\mathrm{var}(X_i) < \infty$.

Definition A.1. (m-Dependence [64, Definition 1]) If for some constant m with $s - r > m$ two sets $(X_1, X_2, ..., X_r)$, $(X_s, X_{s+1}, ..., X_n)$ are independent, then Y^n is said to be *m-dependent*.

We further set

$$\varsigma_i^2 = \mathrm{var}(X_{i+m}) + 2\sum_{j=1}^{m} \mathrm{cov}(X_{i+m-j},\, X_{i+m}), \qquad \text{for } i = 1, 2, ... \tag{A.17}$$

and define

Definition A.2. (Central Limit Theorem (CTL) for Dependent Random Variables [64, Theorem 1])
Let Y^n be a m-dependent sequence of random variables such that
(a) $\mathbb{E}[X_i] = 0$, $\mathbb{E}[|X_i|^3] < \infty$ $(i = 1, 2, ...)$
(b) $\lim_{p\to\infty} \frac{1}{p}\sum_{h=1}^{p} \varsigma_{i+h}^2 = \varsigma^2$ exists, uniformly for all $i = 0, 1, ...$
Then as $n \to \infty$ the random variable $\frac{1}{\sqrt{n}}(X_1 + ... + X_n)$ has a limiting normal distribution with mean 0 and variance ς^2.

Continuing with the DoS of the inflated manifold. We change Equation 3.11 for ease of readability.

$$Z = \sum_{i=1}^{D-d} \underbrace{\log p(E_i)}_{N_i} + \sum_{i=1}^{d} \underbrace{\log p_{\mathcal{N}}(g_i(M))}_{M_i} \underbrace{- \frac{1}{2} \log |\det G_g(M)|}_{G_M} \tag{A.18}$$

$$= \sum_{i=1}^{D-d} N_i + \sum_{i=1}^{d} M_i + G_M. \tag{A.19}$$

For the CTL for dependent random variables we need all random variables to be centered. Thus we set

$$Z' = \sum_{i=1}^{D-d} (N_i - \mathbb{E}[N_i]) + \sum_{i=1}^{d} (M_i - \mathbb{E}[M_i]) + (G_M - \mathbb{E}[G_M]) \tag{A.20}$$

$$= \sum_{i=1}^{D-d} N_i' + \sum_{i=1}^{d} M_i' + G_M'. \tag{A.21}$$

Note that VP and NVP normalizing flows can change the location of their DoS arbitrarily (Corollary 3.2), so the above adjustment could be made by the normalizing flows and does not hinder some of our hypothesis.

The Sequence A.16 then becomes

$$Y^n = Y^{D+1} = \{\underbrace{M_1',\, M_2',\, ...,\, M_d',\, G_M'}_{d+1},\, N_1',\, ...,\, N_{D-d}'\}. \tag{A.22}$$

Therefore, Y^n is $(d+1)$-dependent, since the noise N_1', ..., N_{D-d}' is independent of the manifold part.

For Condition (a) of the CTL for dependent random variable is: $\mathbb{E}[X_i] = 0$, and $\mathbb{E}[|X_i|^3] < \infty$. $\mathbb{E}[X_i] = 0$ holds by construction of Z' (Equation A.21). By sampling, $\mathbb{E}[|M_i'|^3] \approx 1.08$, which is smaller ∞. That $\mathbb{E}[|N_i'|^3] < \infty$ must be ensured by the choice of noise. For the case of normal noise, N_i' and M_i' follow the same distribution and thus this condition holds[1]. $\mathbb{E}[|G_M'|^3] < \infty$ is a constraint on the manifold, which can be interpreted as follows: the logged and centered volume change from the latent d-dimensional normal distribution to the manifold should not be to extreme. Or, in other words: there should be no areas of significant probability in the latent distribution of the manifold that are compressed to a point or stretched to an area of infinite space on the manifold.

Condition (b) is $\lim_{p\to\infty} \frac{1}{p} \sum_{h=1}^{p} \varsigma_{i+h}^2 = \varsigma^2$ exists, uniformly,
with $\varsigma_i^2 = \mathrm{var}(X_{i+m}) + 2 \sum_{j=1}^{m} \mathrm{cov}(X_{i+m-j},\, X_{i+m})$.
Note that $m = d+1$ and thus $X_{i+m} = X_{i+d+1} = N_i'$ comes from a noise variable. Moreover, there is no covariance between any variable and the noise variables. Thus, $\varsigma_i^2 = \mathrm{var}(N_i') =: \varsigma^2$ for all $i = 1, 2, ...$ and thus condition (b) holds.

Then, when $D \to \infty$, $\frac{d}{D} \to 0$ and $\frac{1}{\sqrt{D+1}} Z'$ has a limiting normal distribution with mean 0 and variance ς^2, where ς^2 is determined by the type of the noise.
□

[1] N_i and M_i are the DoS of a normal distribution. Differing variance of normal distribution and of the M_i can be neglected as variance only translates the DoS (Lemma 3.2) and we consider the centered version of N_i and M_i.

A.1.8 Sampling the DoS along the Learned Manifold

Standard normalizing flows cannot sample along the learned manifold. Therefore, we cannot directly sample the DoS of the learned manifold (see Section 3.1.1 on how to sample the DoS).

To get around this, we use the density q_M along the manifold (see Section 2.5.2) and evaluate q_M at points with equal intervals δ along $\mathcal{M}$. This gives an approximation of the probability for an interval $(\delta \cdot q_M(m))$. From this discretized version of q_M, we sample using inverse transform sampling and use these samples to sample the DoS using the method from Section 3.1.1. Obviously, the difference between the discretized and continuous versions of q_M vanishes when $\delta \to 0$.

A.1.9 Smoothed Distributions do not have degenerate-DoS

In this section we show that if a distribution p_M has unbounded peaks in its DoS, smoothing this distribution with non-degenerate-DoS noise erases these peaks.

Let p_M have a DoS with unbounded peaks. Let $D_M = \log p_M(M)$ be random variable distributed according to the DoS of p_M. The peaks occur because there exists a $c \in \mathbb{R}^+$ such that $P[p_M(M) = c] = P[D_M = \log c] > 0$. That is, the DoS of p_M is (partially) discrete (for $D_M = \log c$). Informally, this happens because the density $p_M(\cdot)$ has one density level c, which occurs on an area that has a volume (think of the density of a uniform distribution).

We smooth p_M with non-degenerate-DoS noise E (e.g. normal noise) such that p_{M+E} becomes the distribution of interest.

The pdf of p_{M+E} can be computed by convolution like

$$(p_M * p_E)(x) = \int_{-\infty}^{\infty} p_M(s)\, p_E(x-s)\, ds. \tag{A.23}$$

We split $p_M(\cdot)$ into parts where $p_M(x) = c$ and where $p_M(x) \neq c$ to obtain a clearer expression for $(p_M * p_E)$.
Let S be the set of points for which $p_M(s) = c$, i.e. $S = \{s \,|\, p_M(s) = d : \forall s \in \mathbb{R}\}$, then

$$\begin{align}
(p_M * p_E)(x) &= \int_{-\infty}^{\infty} p_M(s)\, p_E(x-s)\, ds \tag{A.24}\\
&= \int_S p_M(s)\, p_E(x-s)\, ds + \int_{\mathbb{R}\setminus S} p_M(s)\, p_E(x-s)\, ds \tag{A.25}\\
&= \int_S d\, p_E(x-s)\, ds + \int_{\mathbb{R}\setminus S} p_M(s)\, p_E(x-s)\, ds \tag{A.26}\\
&= d \underbrace{\int_S p_E(x-s)\, ds}_{\alpha} + \underbrace{\int_{\mathbb{R}\setminus S} p_M(s)\, p_E(x-s)\, ds}_{\beta} \tag{A.27}
\end{align}$$

Due to the complexity of α and β, almost surely $P[p_{M+E}(M+E) = c] = 0$[1] and thus the DoS of p_{M+E} does not have unbounded peaks.

This proof easily extend to multiple discrete DoS points of p_M.

□

[1] M and E need to have some codependent distributions which is highly unlikely.

A.1.10 Smoothing Changes the DoS

Let $M \sim p_M$ be the density of the manifold, which we smooth with noise $E \sim p_E$, such that only the manifold density and its tangent space is influences. Clearly, the resulting distribution is smoothed and has some new density p_{M+E}. Since $p_{M+E} \neq p_M$, the DoS which is the density of $\log p_{M+E}(M+E)$ and of $\log p_M(M)$ are not alike, and thus smoothing changes the DoS.

A.2 RealVP is more Expressive than NICE

We prove that RealVP is more expressive than NICE [1] by creating a transformation that NICE cannot express but RealVP can, and by showing that RealVP can express all transformations of NICE.
Let a normalizing flow $f = f_s \circ f_c : \mathcal{X} \to \mathcal{Z}$ consist of one coupling layer f_c and one scale layer f_s.
The input to the normalizing flow is $x = (a, b, c)$ such that $x_{\text{id}} = (a),\ x_{\text{change}} = (b, c)$.
We use the superscript ^{N} for a variable or function of NICE and ^{R} for RealVP, e.g. t^N.

The transformation for NICE is

$$f_{\text{NICE}} : (a, b, c) \mapsto \begin{pmatrix} \varsigma_1^N \cdot a \\ \varsigma_2^N \cdot (b + t_1^N(a)) \\ \varsigma_3^N \cdot (c + t_2^N(a)) \end{pmatrix}^T , \tag{A.28}$$

where $\varsigma_1^N, \varsigma_2^N, \varsigma_3^N$ are the scalings from the scale layer.

The transformation for RealVP is:

$$f_{\text{RealVP}} : (a, b, c) \mapsto \begin{pmatrix} \varsigma_1^R \cdot a \\ \varsigma_2^R \cdot \phi_1(a) \cdot (b + t_1^R(a)) \\ \varsigma_3^R \cdot \phi_2(a) \cdot (c + t_2^R(a)) \end{pmatrix}^T , \tag{A.29}$$

where $\phi_i(a) = \frac{\exp(s_i^R(a))}{\nu(a)}$ is the scaling from the scale net (see Equation 2.12) and $\varsigma_1^R, \varsigma_2^R, \varsigma_3^R$ are the scalings from the scale layer.

There is a transformation in the function space of RealVP that is not in that of NICE
For NICE to be equally expressive $f_{\text{NICE}}(x) = f_{\text{RealVP}}(x)$ should hold for all choices of the scaling s^R of RealVP.

Let RealVP choose

$$s_1^R(a) = a \;\text{ and }\; s_2^R(a) = 2a$$

then

$$\phi_1(a) = \frac{\exp(a)}{\sqrt{\exp(3a)}} = \exp\left(-\frac{1}{2}a\right) \;\text{ and }\; \phi_2(a) = \frac{\exp(a)}{\sqrt{\exp(3a)}} = \exp\left(\frac{1}{2}a\right).$$

We evaluate the second dimension of the transformation.

$$
\begin{aligned}
f_{\text{NICE}}(x)_2 &\stackrel{?}{=} f_{\text{RealVP}}(x)_2 && && \text{(A.30)}\\
\varsigma_2^N\,(b + t_1^N(a)) &= \varsigma_2^R\,\phi_1(a)\,(b + t_1^R(a)) && \text{omitting indices to avoid clutter} && \text{(A.31)}\\
& && \text{set } \varsigma^N = \varsigma^R && \text{(A.32)}\\
b + t^N(a) &= \phi(a)\,b + \phi(a)\,t^R(a) && \text{set } t^N(a) = \phi(a)\,t^R(a) && \text{(A.33)}\\
b &= \phi(a)\,b && && \text{(A.34)}\\
1 &\neq \phi(a) && && \text{(A.35)}
\end{aligned}
$$

Thus there exist a transformation in the function space of RealVP that does not exist in that of NICE and thus RealVP is more expressive.

Therefore, when setting $\varsigma^R = 1$, a simple random variable that RealVP can express, but NICE cannot is for instance

$$\left(A,\ B\,e^{-\frac{1}{2}A},\ C\,e^{\frac{1}{2}A}\right),$$

where A, B, C are i.i.d. distributed with the used latent distribution.

All functions in the functions space of NICE are in that of RealVP
If we set $s^R(\cdot) = 0$, then $\phi(\cdot) = 1$ and VP affine coupling layers have the same form as additive coupling layers and are thus identical.
□

A.3 Model and Training

Model
For the cases where $D > 2$, we cloned the RealNVP implementation from `https://github.com/PolinaKirichenko/flows_ood`, used in [34]. We use 10 coupling layers, the *st* net (Equation 2.6) respectively the t net (Equation 2.9) consists of 4 fully connected hidden layers, each consisting of $2D$ hidden neurons. For the $D = 2$, case we used our own implementation, with 16 hidden neurons. We use a rectified linear unit (ReLu) as activation function.

Training
We use a batch size of 32, and the Adam optimizer [65] with a learning rate of 10^{-3} and a weight decay of 5×10^{-5}. With the exception of the learning rate, these settings were chosen following [34].

Our training set contains 2000 samples[1] and our test set contains 1000 samples. We evaluate the test set every 10 epochs. When the average likelihood has improved, we save the model. We stop training when the training loss has not improved for 100 epochs or 5000 epochs are reached. For the train loss see Figure 4.15.

[1]Considering the curse of dimensionality and that one digit of MNIST has 10,000 samples and an intrinsic-dimensionality of about 10 [54], 2000 samples for a distribution with a 1 dimensional manifold should be more than sufficient to approximate the density.

We use 8000 samples for the evaluation.
The full implementation can be found at `https://github.com/luifire/master_thesis`.

Due to computational constraints (we need to train $3 \times 3 \times 4 \times 5 = 180$ models), we do not perform cross-validation. However, we compensate for this by using many different datasets for which we obtain consistent results.

A.4 Smooth Step Distribution

We created the smooth step distribution to obtain a distribution with a special DoS, that is two Dirac points with unequal mass and a smooth function in between these points.
Its pdf is

$$p_{\text{Smooth Step}}(x) = \begin{cases} 0 & \text{for } x < s_{\text{start}} \text{ or } x > s_{\text{end}} \\ \frac{1}{Z}(s_{\text{range}} \cdot \psi(s_{\text{growth}} \cdot x) + s_{\text{offset}}) & \text{else} \end{cases} \tag{A.36}$$

$$\psi(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ 1 & \text{for } x \geq 1 \\ x^2 \cdot (3 - 2x) & \text{else} \end{cases} \tag{A.37}$$

$$Z = \int_{s_{\text{start}}}^{s_{\text{end}}} s_{\text{range}} \cdot \psi(s_{\text{growth}} \cdot x) + s_{\text{offset}} \, dx \tag{A.38}$$

$$s_{\text{range}} = s_{\text{end}} - s_{\text{start}}. \tag{A.39}$$

We choose $s_{\text{start}} = -5, s_{\text{end}} = 5, s_{\text{growth}} = 0.5, s_{\text{offset}} = 1$. For the resulting DoS and pdf see Figure A.2, respectively Figure 4.1.

A.5 DoS's of the Manifolds

The DoS's of the manifold distributions are shown in Figure A.2.

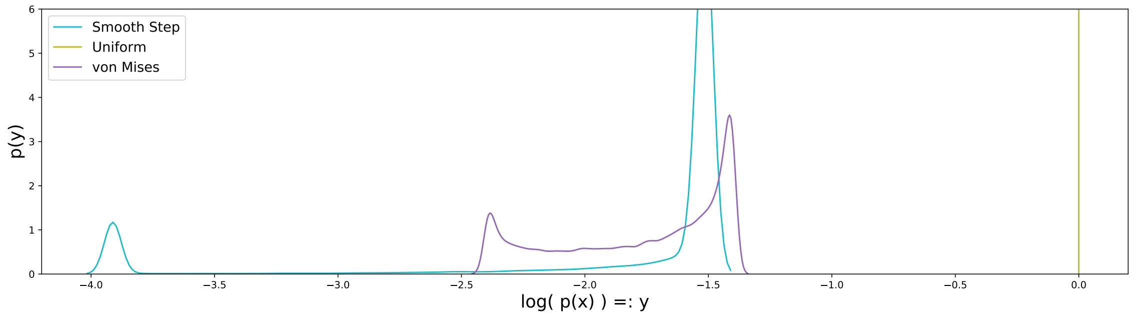


Figure A.2: Kernel density estimations of the DoS of the manifold without added noise. The DoS of the the Smooth Step distribution is expressed by two Dirac delta function terms of different mass and a connection between these points. We visualize the unbounded peaks with two bumps containing the respective masses. The DoS of the uniform distribution expressed by a Dirac delta function. The DoS of the von Mises has Dirac .

A.6 All Experimental Results

In this section we again show all divergences, learned pdfs and DoS's for all manifold from the experiments in Section 4.2. We leave out the caption of the figure s.t. they fit on one page.

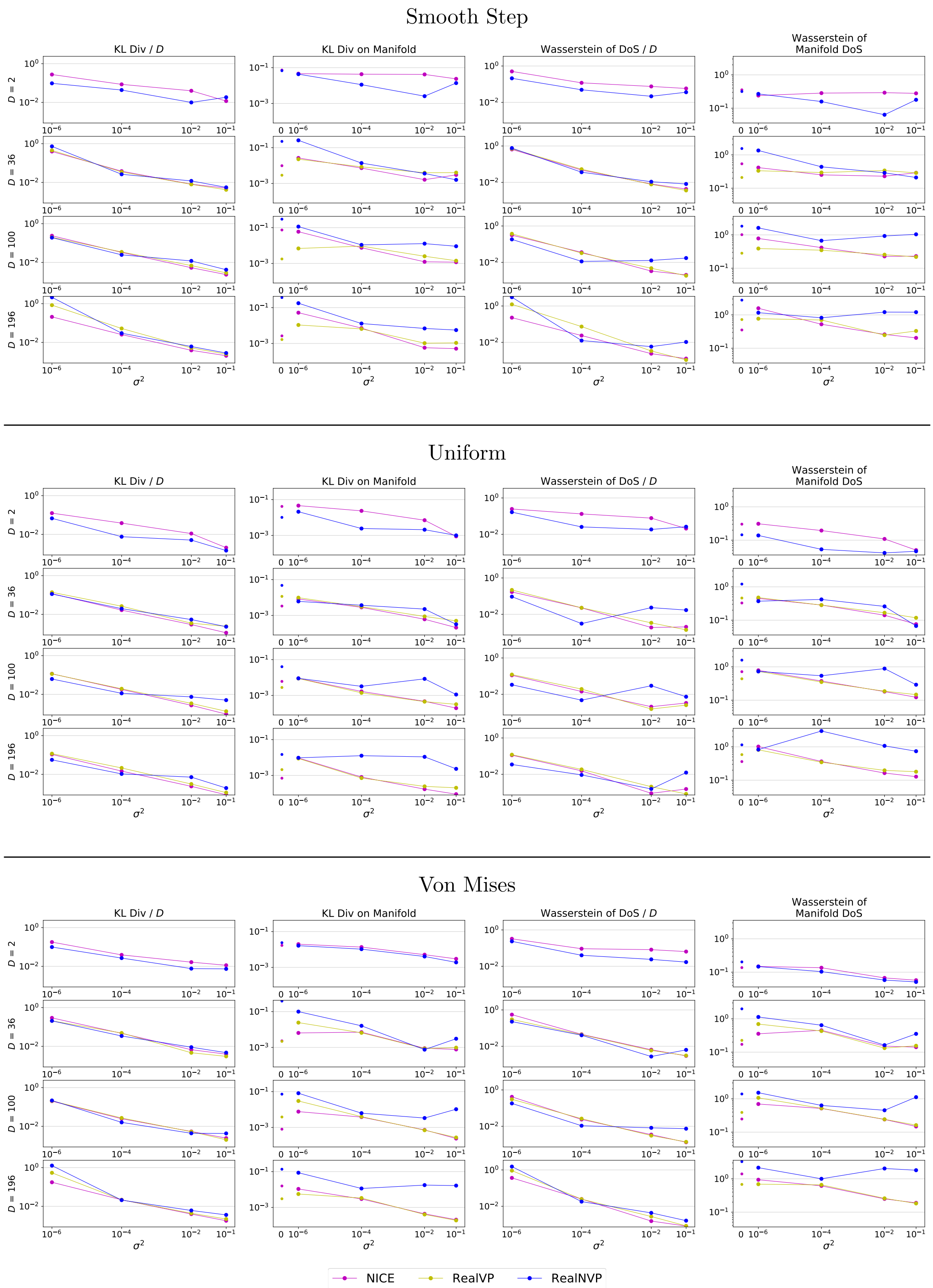


Figure A.3: All divergences measured of all models, dimensions and variances. Rows show the embedding dimension D, and the x-axes show the noise strength used for inflation. Both axes are in log-scale. Let p_X be the distribution of the inflated manifold and q_X a model (see legend) trained on samples of p_X. Let q_M be the distribution of q_X along the ground truth manifold p_M. "KL Div / D" is the KL divergence from q_X to p_X. "KL Div on Manifold" is the KL divergence from q_M to p_M (along the manifold). "Wasserstein of DoS / D" is between the DoS of p_X and q_X and "Wasserstein of Manifold DoS" between the DoS of q_M and p_M. For $D = 2$, RealVP coincides with NICE.

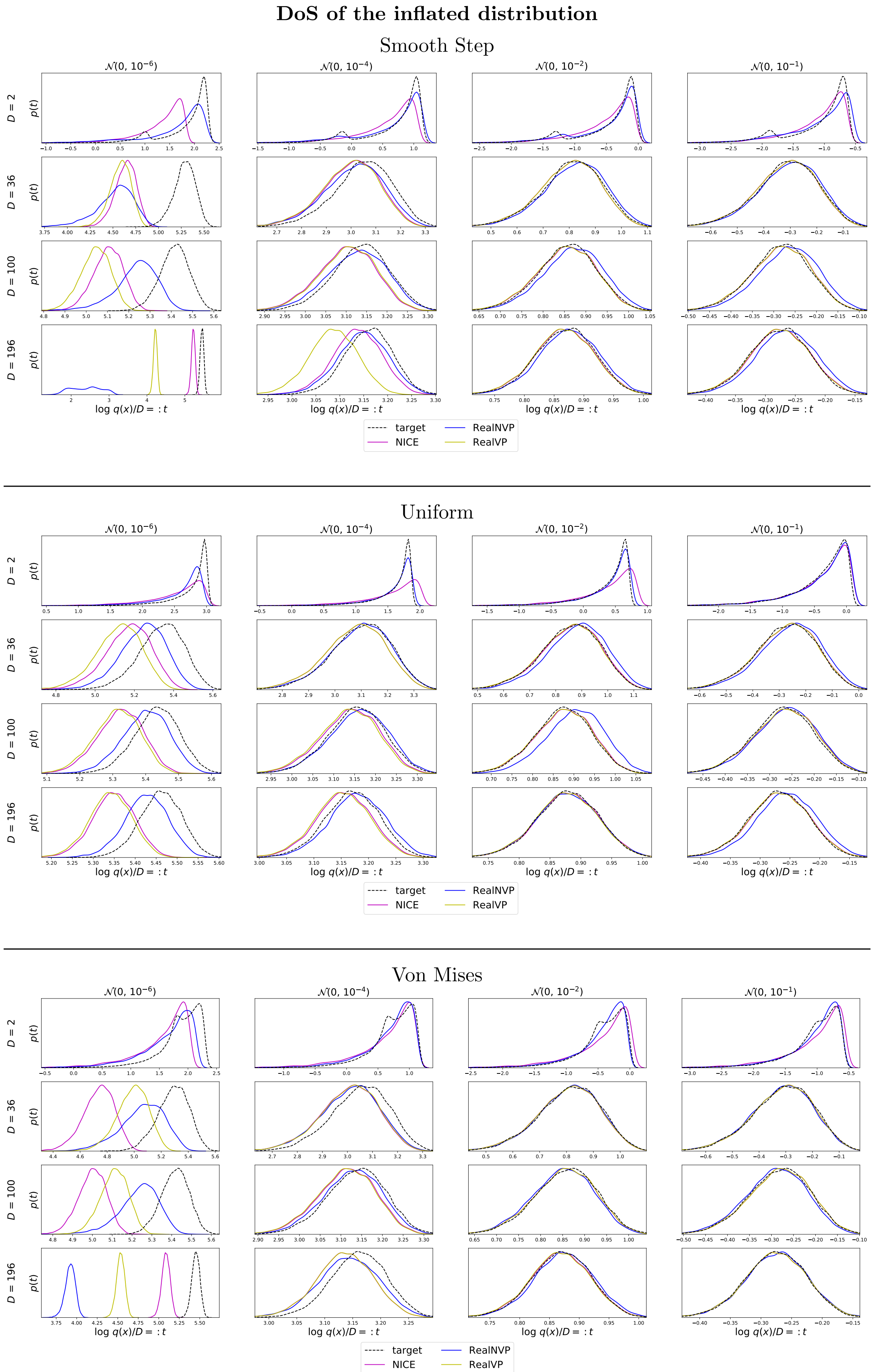


Figure A.4: See Figure 4.3.

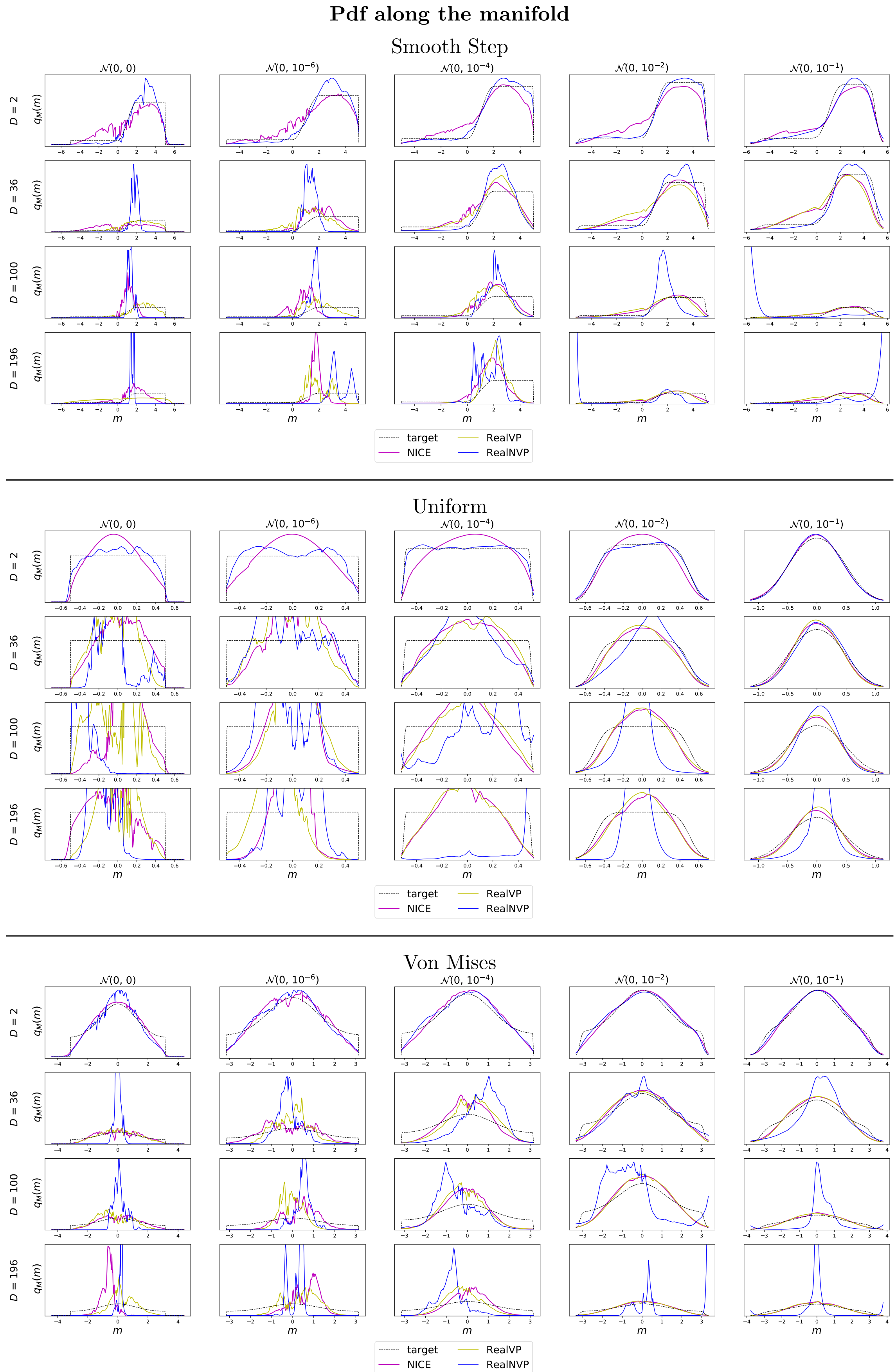
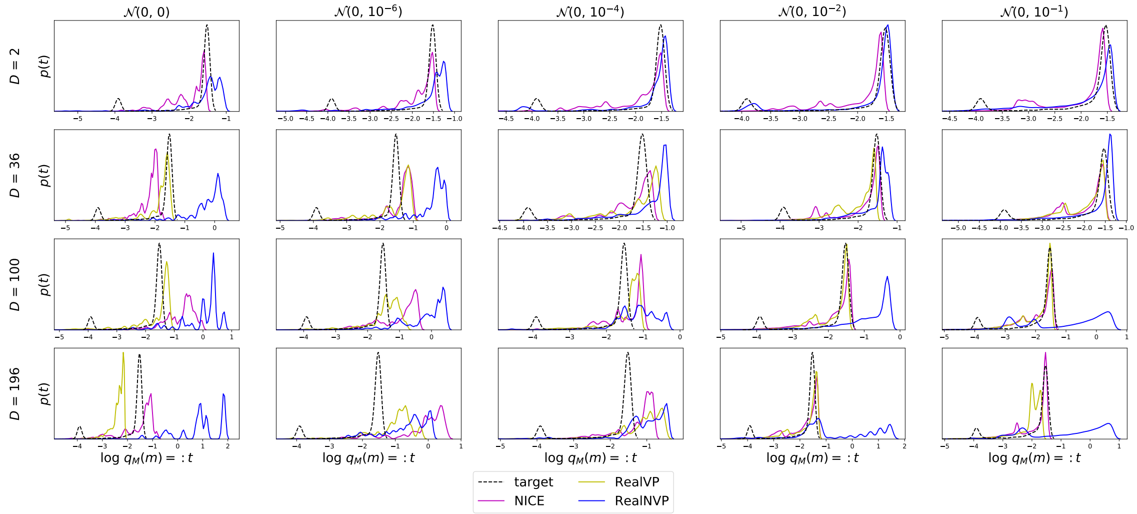


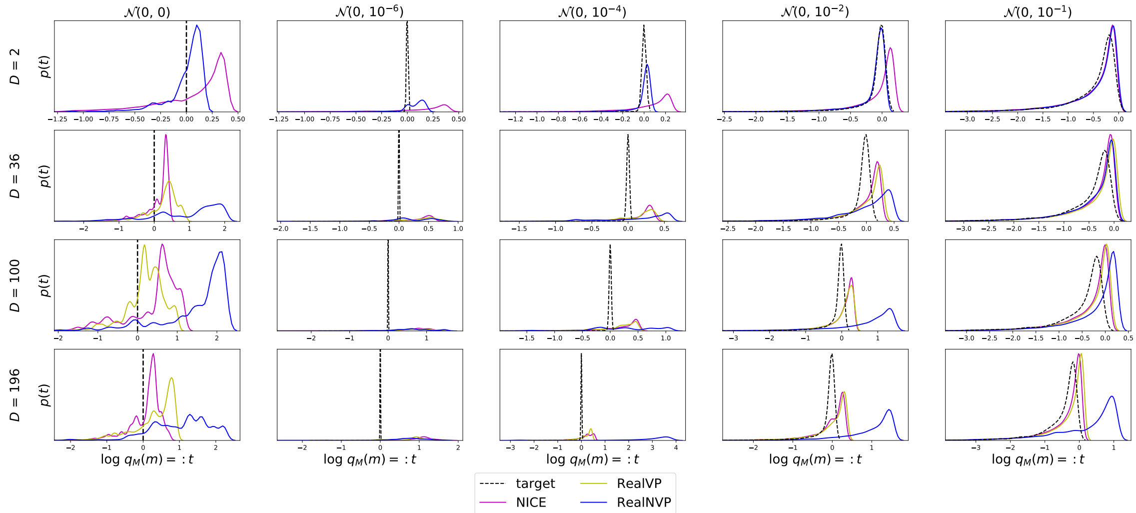
Figure A.5: See Figure 4.6

DoS of the manifold

Smooth Step



Uniform



Von Mises

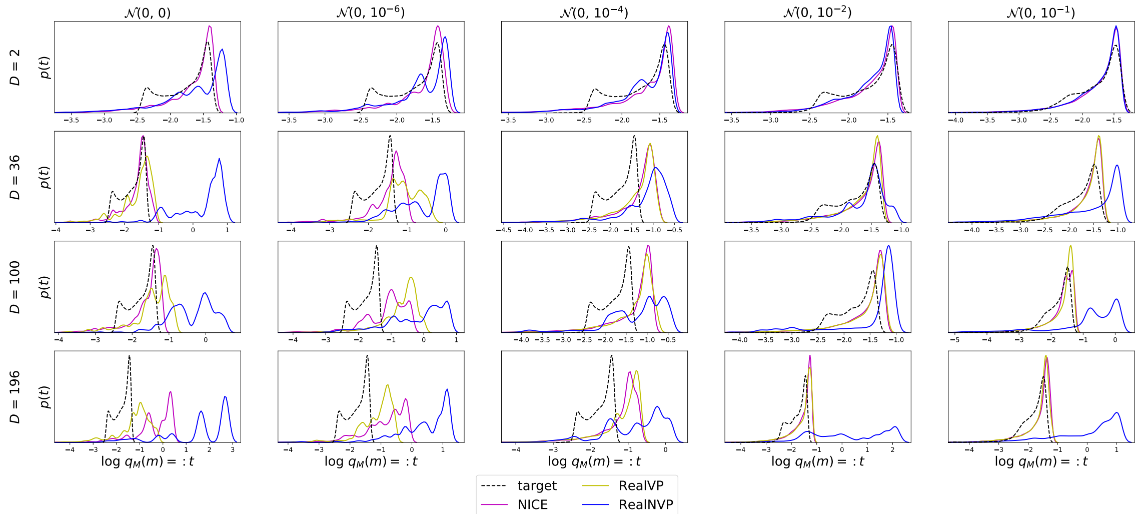


Figure A.6: See Figure 4.8

Uniform and normal noise comparison